

Entropy as Origin: Obidi's Theory of Entropicity and the Informational Genesis of Matter and Spacetime

A Theoretical Physics Monograph

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June 2026

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Submitted for scholarly consideration and open review

Abstract

This monograph presents *Obidi's Theory of Entropicity* (ToE) — an original theoretical framework proposing that the entirety of physical reality, encompassing matter, mass-energy, quantum fields, and the geometric structure of spacetime, emerges from the entropic dynamics of an underlying information-geometric substrate. The theory is grounded in the recognition that information geometry — the mathematical discipline studying families of probability distributions as Riemannian manifolds equipped with the Fisher information metric — provides the most natural and rigorous ontological

foundation for physical existence. Where previous approaches to information-theoretic physics have treated information as a powerful descriptive analogy for physical phenomena, Obidi's Theory of Entropicity claims that information geometry is the literal, constitutive substrate of physical reality, and that all physical phenomena are projections, condensations, or gradient manifestations of the geometry and dynamics of this informational manifold.

The pivotal conceptual and mathematical fulcrum of the theory is Landauer's Principle — the thermodynamic result, established by Rolf Landauer in 1961, that the irreversible erasure of one bit of information necessarily dissipates a minimum energy of $E = kT \ln 2$ into the environment. In Obidi's Theory of Entropicity, Landauer's Principle is elevated from a result of computational thermodynamics to an ontological mechanism of cosmological scope. The erasure of a bit at the level of the informational substrate is not merely accompanied by energy dissipation — it *is* the generative act by which abstract informational potential is converted into concrete physical actuality. The energy released by bit erasure — the Landauer quantum — is identified as the primordial form of physical energy from which all mass-energy in the observable universe derives.

The theory introduces the concept of *Entropicity* — defined as the intrinsic tendency of information-geometric configurations to undergo irreversible resolution, collapsing from high-entropy superpositions of informational states into actualized, lower-entropy physical configurations. The Entropicity Field, governed by a Laplace-Beltrami field equation over the statistical manifold equipped with the Fisher metric, drives clusters of correlated bit erasure events — termed *entropic condensation events* — which produce localized concentrations of Landauer energy identifiable with fundamental particles. The geometry of erasure patterns across the statistical manifold gives rise to the metric tensor of emergent spacetime, providing a natural derivation of gravitational curvature as an entropic gradient. The theory demonstrates that the second law of thermodynamics, quantum measurement

(wave-function collapse), the Bekenstein-Hawking black hole entropy formula, the holographic principle, the cosmological constant, and quantum entanglement all find unified, natural explanations within the ToE framework. Empirical predictions — including modified dispersion relations for high-energy photons, Planck-scale thermodynamic corrections, and entropic corrections to galactic dynamics — are presented and discussed. The Theory of Entropicity thus offers a candidate unified framework reconciling quantum mechanics, thermodynamics, and general relativity at the level of their shared informational foundation.

Keywords: Information geometry, Landauer's Principle, Theory of Entropicity, emergence of matter, entropic gravity, spacetime curvature, Fisher information metric, holographic principle, bit erasure, thermodynamic irreversibility

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1. Introduction: The Information-Theoretic Turn in Fundamental Physics

Physics has undergone, in the course of the past century, a succession of conceptual revolutions each of which has demanded a radical reconceptualization of the nature of physical reality. The Newtonian synthesis gave way to statistical mechanics and thermodynamics, which revealed that the macroscopic properties of matter — temperature, pressure, entropy — are not fundamental but rather statistical emergents of the collective behavior of vast numbers of microscopic constituents. The quantum revolution then demolished the classical picture of particles as localized

objects with definite properties, replacing it with wavefunctions, superpositions, and probabilistic measurement outcomes. General relativity, meanwhile, reconceived gravity not as a force but as the geometry of a four-dimensional spacetime manifold, curved by the presence of mass and energy. Each of these revolutions has involved a transition from a more concrete and intuitive ontology to a more abstract and mathematical one.

The present epoch of theoretical physics appears poised at the threshold of a further revolution, arguably the most radical yet: the proposition that information itself — not matter, not energy, not spacetime, but abstract *information* — is the most fundamental ingredient of physical reality. This is not a metaphor or a computational convenience. It is a substantive ontological claim, supported by a growing convergence of independent lines of theoretical and experimental inquiry, each pointing toward the same conclusion: physical reality is, at its deepest level, informational in nature.

The seeds of this information-theoretic turn were planted in 1948 when Claude Shannon published his foundational paper on the mathematical theory of communication, establishing that information could be quantified rigorously in terms of entropy — the same mathematical function, up to a constant, that Boltzmann had used to characterize thermodynamic disorder. Shannon's entropy, $H = -\sum p_i \log p_i$, is formally identical to Boltzmann's thermodynamic entropy $S = -k \sum p_i \ln p_i$, a coincidence that was quickly recognized as more than superficial. If the mathematics of information theory and the mathematics of thermodynamics are the same, perhaps their physical content is also the same. Perhaps the entropy that increases in irreversible thermodynamic processes and the information that is lost in noisy communication channels are not merely analogous — perhaps they are identical.

The physicist most responsible for elevating information to a cosmological principle was John Archibald Wheeler, who in his later years articulated the thesis that he summarized in the memorable slogan "It from Bit." For Wheeler, every particle, every field, every spacetime event — every "it" — derives its existence from yes-or-no answers to binary questions — "bits." The physical world, in Wheeler's vision, is not made of matter and energy in any primary sense; it is made of information, and matter and energy are the way that information manifests when observed and recorded by physical systems. Wheeler's thesis was more of a research program than a theory — a philosophical orientation rather than a set of equations — but it set the agenda for a generation of physicists who would seek to give precise mathematical form to the intuition that information is physically primary.

The most concrete and quantitatively precise evidence for the physical primacy of information came from the study of black holes. In 1972–1973, Jacob Bekenstein, reasoning from thermodynamic analogies and quantum mechanical considerations, proposed that black holes possess an entropy proportional to the area of their event horizon, not their volume: $S_{\text{BH}} = A / 4l_{\text{P}}^2$, where A is the horizon area and l_{P} is the Planck length. Stephen Hawking's subsequent derivation of the Hawking temperature and the thermal radiation of black holes confirmed and quantified Bekenstein's proposal, placing it on firm theoretical ground. The Bekenstein-Hawking entropy formula has profound implications: it implies that the maximum information content of any region of space is proportional to its boundary area, not its volume. This is the holographic principle — the idea that three-dimensional physical reality is, in some deep sense, encoded on a two-dimensional surface, just as a hologram encodes three-dimensional information on a flat plate.

More recently, Erik Verlinde demonstrated in 2011 that Newton's law of gravity — and, with greater mathematical effort, Einstein's field equations of

general relativity — can be derived from entropic considerations alone, treating gravity not as a fundamental force but as an entropic gradient: the thermodynamic tendency of a system to move toward states of higher entropy. Verlinde's entropic gravity framework, while still controversial and incomplete as a full theory of quantum gravity, represents a major step toward the vision of spacetime and gravity as emergent, thermodynamic phenomena rather than fundamental constituents of reality.

It is against this background — Shannon's entropic information, Wheeler's It from Bit, Bekenstein and Hawking's black hole thermodynamics, the holographic principle, and Verlinde's entropic gravity — that *Obidi's Theory of Entropicity* (hereafter ToE) is to be understood. ToE is not merely another contribution to this tradition; it is a coherent synthesis and original extension that addresses the central lacuna of all previous approaches: the failure to specify a *physical mechanism* by which information becomes matter. All previous information-theoretic approaches to physics have demonstrated that information and matter are deeply related, that information-theoretic quantities appear in physical equations, that entropic reasoning can reproduce gravitational dynamics. None has provided an account of how and why the transition from information to physical reality occurs — what the generative act is by which abstract information becomes concrete matter.

Obidi's Theory of Entropicity provides exactly this missing mechanism. The theory identifies Landauer's Principle — the thermodynamic principle, established by Rolf Landauer in 1961, that the irreversible erasure of one bit of information dissipates a minimum energy of $E = kT \ln 2$ — as the ontological bridge between the informational substrate and the physical world. In ToE, Landauer's Principle is reinterpreted not as a result of computational thermodynamics but as the fundamental mechanism of physical genesis: the act by which bits are erased at the level of the

informational substrate is the act by which physical reality is created. The energy released by bit erasure — what the present monograph will call the *Landauer quantum* — is not merely a computational cost; it is the primordial energy from which all mass and force in the observable universe derives.

The central thesis of this monograph may be stated precisely and without ambiguity: *the irreversible erasure of information at the Planck-scale informational substrate — a pre-geometric statistical manifold equipped with the Fisher information metric — is the primordial physical act from which the universe's material and geometric structure arises. Physical reality is the ongoing process of informational foreclosure, and the Theory of Entropicity is the mathematical and philosophical framework that describes this process in full detail.*

The structure of the argument is as follows. Section 2 establishes the mathematical foundations of information geometry, with particular attention to the Fisher information metric and the geometry of statistical manifolds — the mathematical infrastructure of the informational substrate. Section 3 presents Landauer's Principle in full rigor, traces its history and experimental confirmation, and then executes the central reinterpretation that is the theoretical core of ToE. Section 4 develops the full framework of the Theory of Entropicity, including the definition of Entropicity, the structure of the informational substrate, the emergence mechanism, the Entropicity field equations, and the derivation of mass as informational inertia. Section 5 addresses the emergence of spacetime and the derivation of gravity as an entropic gradient within the ToE framework. Section 6 connects ToE to established physics, showing how the second law of thermodynamics, quantum measurement, black hole thermodynamics, dark energy, and quantum entanglement all find natural homes within the theory. Section 7 explores the philosophical implications. Section 8 presents empirical

predictions. Sections 9 and 10 provide the mathematical appendix and a discussion of open questions. Section 11 concludes.

2. Foundations: What Is Information Geometry?

Information geometry is the discipline that studies the geometric properties of families of probability distributions, treating them as the points of a differentiable manifold and equipping that manifold with a natural metric derived from the statistical structure of the distributions themselves. The discipline was systematized principally by Shun-ichi Amari, whose work — from his 1985 monograph to his 2016 comprehensive treatment — established information geometry as a rigorous and powerful mathematical framework with applications ranging from statistics and machine learning to neuroscience and physics. For the purposes of Obidi's Theory of Entropicity, information geometry provides the precise mathematical language needed to describe the informational substrate of physical reality. We therefore develop its central concepts in some depth before proceeding to the theory proper.

2.1 Statistical Manifolds and the Fisher Information Metric

Consider a family of probability distributions parameterized by a vector of real-valued parameters $\theta = (\theta^1, \theta^2, \dots, \theta^n) \in \Theta \subseteq \mathbb{R}^n$, where each value of θ specifies a distinct probability distribution $p(x|\theta)$ over a sample space X . The set of all such distributions, $S = \{p(x|\theta) : \theta \in \Theta\}$, forms a *statistical manifold* — a smooth manifold whose points are probability distributions. The coordinates on this manifold are the parameters θ^i , and the geometry of the

manifold is determined by the way in which neighboring probability distributions are statistically related to one another.

The natural metric on a statistical manifold is the *Fisher information metric*, defined by:

$$g_{ij}(\theta) = \mathbb{E}_{x \sim p(\cdot|\theta)} [\partial_i \log p(x|\theta) \cdot \partial_j \log p(x|\theta)] \text{ (Eq. 1) — Fisher Information Metric}$$

where $\partial_i = \partial/\partial\theta^i$ denotes partial differentiation with respect to the i -th parameter, and the expectation is taken over x distributed according to $p(x|\theta)$. Equivalently, in integral form:

$$g_{ij}(\theta) = \int p(x|\theta) [\partial_i \log p(x|\theta)] [\partial_j \log p(x|\theta)] dx \text{ (Eq. 2) — Fisher Metric,}$$

Integral Form

The Fisher information metric is positive semi-definite (and generically positive definite on well-behaved families of distributions), symmetric, and transforms correctly as a tensor under reparameterization of θ . These properties establish it as a genuine Riemannian metric on the statistical manifold. The squared distance $ds^2_F = g_{ij} d\theta^i d\theta^j$ between two infinitesimally separated distributions $p(x|\theta)$ and $p(x|\theta+d\theta)$ measures, in a precise statistical sense, how *distinguishable* the two distributions are: distributions that are statistically indistinguishable (essentially the same) are close in this metric, while distributions that differ markedly in their predictions are far apart.

This statistical distinguishability interpretation of the Fisher metric is of great importance for ToE: it means that the geometry of the statistical manifold encodes the physical distinguishability of configurations. Regions of the manifold where the Fisher metric is large — where nearby distributions are highly distinguishable — correspond, in the physical projection, to regions of high information density and, as we shall see, high curvature of emergent spacetime.

The geodesics of the statistical manifold — the shortest paths between distributions under the Fisher metric — have a beautiful interpretation: they are the most statistically efficient paths between probability distributions in the sense that they minimize the total statistical "cost" (measured by the accumulated Fisher divergence) of moving from one distribution to another. In the physical interpretation of ToE, geodesics of the statistical manifold correspond to the paths of least entropic resistance — the natural trajectories of physical processes in the emergent spacetime.

Remark on Notation

Throughout this monograph, Latin indices ($i, j, k, \dots$) refer to coordinates on the statistical manifold, while Greek indices ($\mu, \nu, \alpha, \beta, \dots$) refer to coordinates on emergent spacetime. The Einstein summation convention is employed unless explicitly stated otherwise. The Fisher metric g_{ij} is always the metric on the statistical manifold; the spacetime metric is written $g_{\mu\nu}$.

2.2 Amari's Dual Connections and the α -Geometry

A Riemannian manifold is characterized not only by its metric but also by its affine connection — the rule for parallel transport and covariant differentiation. In standard Riemannian geometry, there is a unique metric-compatible, torsion-free connection, the Levi-Civita connection. Statistical manifolds, however, possess a richer geometric structure: they admit a family of connections, parameterized by a real number α , that are compatible with the Fisher metric in a generalized sense. This is the starting point of Amari's α -geometry, which fundamentally departs from the standard Riemannian framework.

The two most important connections on a statistical manifold are the *e-connection* (exponential connection, $\alpha = +1$) and the *m-connection* (mixture connection, $\alpha = -1$). The e-connection governs parallel transport in the space of exponential families of distributions — the distributions of the form $p(x|\theta) = \exp[C(x) + \theta^i F_i(x) - \psi(\theta)]$, which include the Gaussian, exponential, Poisson, and many other standard distributions. The m-connection governs parallel transport in the space of mixture families. These two connections are *dual* with respect to the Fisher metric in the following precise sense: for any three vector fields U, V, W on the statistical manifold,

$$W(g(U, V)) = g(\nabla^{(e)}_W U, V) + g(U, \nabla^{(m)}_W V) \text{ (Eq. 3) — Duality of e- and m-connections}$$

where $\nabla^{(e)}$ is the e-connection and $\nabla^{(m)}$ is the m-connection. The general α -connection $\nabla^{(\alpha)}$ interpolates between these two extremes, with $\nabla^{(0)} = \nabla^{\text{LC}}$ being the Levi-Civita connection, and $\nabla^{(\alpha)}$ and $\nabla^{(-\alpha)}$ forming a dual pair for each value of α .

The significance of dual connections for ToE is profound. In the informational substrate, the e-geometry governs the structure of possible configurations of the substrate — how exponential families of probability distributions (encoding information about physical degrees of freedom) are organized. The m-geometry governs the dynamics of the Entropicity field — how mixtures of configurations evolve as entropic resolution proceeds. The interplay between these dual geometric structures is responsible for the rich phenomenology of the physical world that emerges from the substrate.

The general α -curvature tensors — the curvature tensors of the α -connections — encode information about the global structure of the statistical manifold that goes beyond the local Fisher metric. In the ToE framework, these curvature tensors are related to the topological structure of the universe and

to conserved quantities in the physical domain. The precise relationship between α -curvature and physical topology is an important open question that is addressed further in Section 10.

A particularly important property of certain statistical manifolds — exponential families in particular — is *dual flatness*: the vanishing of both the e-curvature and the m-curvature, even though the manifold is generally curved with respect to the Levi-Civita connection. A dually flat manifold admits a remarkable coordinate system — called a *Bregman divergence structure* — in which the Fisher metric takes a particularly simple form and geodesics in both connections are straight lines. The dually flat structure is of direct relevance for ToE's treatment of free (non-interacting) fields, which correspond to exponential-family regions of the statistical manifold.

2.3 Curvature of Statistical Manifolds

The curvature of a statistical manifold — measured by the Riemann curvature tensor R^k_{ijl} of the Fisher metric, or by the scalar curvature R obtained by contraction — has a direct statistical interpretation: it measures the degree to which the statistical manifold fails to be locally Euclidean, or equivalently, the degree to which the geometry of the family of distributions differs from that of independent distributions.

In precise terms: a flat statistical manifold (zero Fisher curvature) corresponds to a family of distributions in which the parameters θ^i are statistically independent of one another — the components of the informational configuration are uncorrelated. A curved statistical manifold corresponds to a family in which the parameters are statistically coupled — the components interact. High curvature in the Fisher metric space therefore reflects the complexity and density of statistical correlations among the components of the informational configuration.

In the physical interpretation of ToE, the curvature of the statistical manifold in the neighborhood of a given configuration encodes the degree to which the informational degrees of freedom in that region are correlated — the degree to which they constitute a coherent physical structure rather than an uncorrelated informational noise. Particles are regions of high Fisher curvature — tightly correlated, highly structured informational configurations. The vacuum is a region of low (but not zero) Fisher curvature — uncorrelated or weakly correlated informational noise, corresponding to the quantum vacuum fluctuations of standard quantum field theory.

The scalar curvature R of the Fisher metric enters naturally into the Entropicity field equation (Section 4.4) as a source term, coupling the intrinsic geometry of the statistical manifold to the dynamics of the Entropicity field. This coupling is the mathematical expression of the physical principle that the structure of the informational substrate influences the dynamics of entropic resolution, which in turn shapes the emergent physical world.

2.4 In ToE: The Informational Substrate as a Statistical Manifold

Having established the mathematical foundations of information geometry, we are now in a position to state the core ontological claim of Obidi's Theory of Entropicity with full precision. The claim is this: the universe's deepest layer of reality — the ontological ground from which all physical phenomena derive — is a statistical manifold, an information-geometric space of possible configurations, equipped with the Fisher information metric as its intrinsic geometry. Spacetime, matter, energy, and force are not fundamental constituents of this substrate; they are derived, emergent structures that

arise when the substrate undergoes the process of entropic resolution (Entropicity).

The statistical manifold that constitutes the informational substrate is of astronomical dimensionality — its parameter space Θ is of dimension comparable to the total number of degrees of freedom in the observable universe, estimated at approximately 10^{122} bits. Each point in this manifold represents a possible configuration of all the informational degrees of freedom of physical reality — a complete specification of the probability distributions over all possible physical states. The physical universe, as we observe it, is not a single point in this manifold but rather a process — a trajectory through the manifold, governed by the Entropicity field equations, along which configurations are progressively actualized by irreversible bit erasure.

Crucially, the informational substrate is *pre-geometric*: it exists prior to and independently of the spacetime that emerges from it. The notion of "prior" here is not temporal (time itself emerges from the substrate) but ontological — the substrate is logically and constitutively more fundamental than spacetime. The Fisher metric of the substrate is not a metric on spacetime; it is a metric on the space of possible configurations, which projects, in the continuum limit and under the appropriate coarse-graining, onto the metric tensor of emergent spacetime. The precise mechanism of this projection is developed in Section 5.

3. Landauer's Principle: The Physical Cost of Forgetting

Of all the conceptual tools that Obidi's Theory of Entropicity brings to bear on the problem of the informational genesis of physical reality, none is more central than Landauer's Principle. This section presents the principle with full rigor, traces its remarkable intellectual history, and then — in what is the central conceptual move of the entire theory — elevates it from a result of computational thermodynamics to an ontological mechanism of cosmological scope.

3.1 Statement and Proof of Landauer's Principle

Landauer's Principle may be stated precisely as follows: any logically irreversible manipulation of information — in particular, the erasure of one bit of information — must dissipate a minimum energy of

$$E_{min} = kT \ln 2 \text{ (Eq. 4) — Landauer's Minimum Erasure Energy}$$

into the environment as heat, where k is Boltzmann's constant ($k \approx 1.38 \times 10^{-23}$ J/K) and T is the absolute temperature of the environment. The corresponding minimum entropy increase of the environment is:

$$\Delta S_{env} \geq k \ln 2 \text{ (Eq. 5) — Minimum Entropy Production per Bit Erasure}$$

The thermodynamic argument establishing this bound proceeds as follows. Consider a one-bit memory — a physical system with two distinguishable states, labeled 0 and 1. Before erasure, the bit may be in either state; the probability distribution over states is $p(0) = p(1) = 1/2$ (maximum entropy state), or more generally, some distribution with entropy $H = -p(0)\log_2 p(0) - p(1)\log_2 p(1) > 0$. Erasure is the operation that takes the bit from *any* initial

state (0 or 1) to a definite final state (say, 0) — a many-to-one logical mapping. This many-to-one character is precisely what makes erasure logically irreversible: one cannot reconstruct the pre-erasure state from the post-erasure state.

From the perspective of statistical mechanics, the initial ensemble of memory states has a statistical entropy of at least $k \ln 2$ (for a one-bit memory). The final state — definite, known — has zero statistical entropy. By the second law of thermodynamics, the total entropy of the system (memory + environment) cannot decrease. Therefore, the reduction in the entropy of the memory must be compensated by an equal or greater increase in the entropy of the environment:

$$\Delta S_{\text{memory}} + \Delta S_{\text{env}} \geq 0 \implies \Delta S_{\text{env}} \geq -\Delta S_{\text{memory}} = k \ln 2 \text{ (Eq. 6) — Entropy Balance for Bit Erasure}$$

If the environment is a thermal reservoir at temperature T , then by the thermodynamic relation $\Delta Q = T \Delta S_{\text{env}}$, the heat dissipated into the environment satisfies:

$$\Delta Q \geq kT \ln 2 \text{ (Eq. 7) — Minimum Heat Dissipation}$$

This is Landauer's bound. Note that the bound is a strict inequality for realistic physical systems — only an idealized, infinitely slow (quasi-static) erasure process could approach the bound — and the excess heat reflects the irreversible entropy production of the actual erasure procedure.

It is worth emphasizing what this argument does and does not assume. It does not assume any specific physical implementation of the memory or the erasure mechanism. It requires only: (a) that information is encoded in distinguishable physical states, (b) that erasure is a logically irreversible operation, and (c) that the second law of thermodynamics applies. The

universality of these assumptions implies the universality of the bound: any physical system that erases information — regardless of whether it is a silicon transistor, a molecular switch, a neural synapse, or a Planck-scale informational degree of freedom — must obey Eq. (7).

3.2 Historical Context and Experimental Validation

The principle bearing Landauer's name was first articulated by Rolf Landauer in his 1961 paper "Irreversibility and Heat Generation in the Computing Process," published in the IBM Journal of Research and Development. Landauer's paper was motivated by a practical concern of computational engineering: what are the fundamental thermodynamic limits on the energy efficiency of computation? His answer — that only logically irreversible operations have unavoidable energy costs — was revolutionary, because it implied its contrapositive: logically reversible computation, in principle, need not dissipate energy at all.

Landauer's work languished in relative obscurity for two decades until Charles Bennett, in his 1982 paper "The Thermodynamics of Computation," used Landauer's Principle to resolve the century-old paradox of Maxwell's Demon. James Clerk Maxwell had proposed, in 1867, a thought experiment in which a tiny intelligent being — the Demon — operates a frictionless door between two chambers of gas, allowing fast molecules to pass in one direction and slow molecules in the other, apparently reducing the entropy of the gas without expending energy, thereby violating the second law.

Bennett's resolution was elegant and definitive. The Demon, in the course of sorting molecules, accumulates information about the molecular velocities in its memory. This information must eventually be erased — otherwise the Demon's memory fills up and it cannot continue operating. The erasure of the Demon's memory, by Landauer's Principle, dissipates at least as much

entropy as was gained by the sorting operation. The second law is saved not by any mechanical friction or quantum uncertainty but by the thermodynamic cost of information erasure. Maxwell's Demon fails precisely because *forgetting costs energy*.

Experimental confirmation of Landauer's bound at the single-bit level was achieved by Antoine Bérut and colleagues in 2012, published in *Nature*. Their experiment used a colloidal particle — a microscopic glass bead — in a double-well potential created by optical traps, representing a two-state memory. By carefully controlling the potential landscape and measuring the heat dissipated during erasure operations at temperatures close to room temperature, Bérut et al. measured a heat dissipation per erasure event consistent with $kT \ln 2$ to within experimental uncertainty. This was the first direct experimental confirmation of Landauer's bound at the single-bit level, and it established beyond reasonable doubt that information erasure has a real, measurable thermodynamic cost — that information is physically real, not merely descriptive.

Subsequent experiments have further refined these measurements. Jun et al. (2014) demonstrated that the Landauer bound can be approached arbitrarily closely (within a few percent) by performing the erasure operation sufficiently slowly. Hong et al. (2016) confirmed the bound in a different physical system, using cold atoms. The consistent experimental confirmation of Landauer's Principle across multiple physical systems and implementations demonstrates that the principle captures a deep, system-independent feature of the relationship between information and physics.

3.3 Landauer's Principle as Ontological Bridge

We come now to the most critical and original step in Obidi's Theory of Entropicity — the step that transforms the entire program from a collection of

suggestive analogies into a coherent physical theory. This step is the ontological reinterpretation of Landauer's Principle.

In the standard context of computational thermodynamics, Landauer's Principle says: when a physical system erases a bit of information, it must transfer at least $kT \ln 2$ of energy to its environment as heat. The bit is erased; the energy flows; the second law is satisfied. In this context, the bit erasure is something that happens *within* physical reality — a computational operation performed by a physical device (logic gate, transistor, molecular switch) embedded in spacetime, operating within the existing framework of thermodynamics. The energy $kT \ln 2$ is dissipated into a thermal reservoir that already exists.

In Obidi's Theory of Entropicity, the situation is radically and fundamentally different. The bit erasures in question occur not within physical reality but *at the level of the informational substrate itself* — at the pre-geometric level of the statistical manifold, prior to the existence of spacetime, prior to the existence of thermal reservoirs, prior to the existence of matter and energy as ordinarily understood. These are not erasures performed by physical devices within the universe; they are the erasures that *constitute* the universe. And the energy released — the Landauer quantum $E = kT \ln 2$ — is not dissipated into a pre-existing thermal environment; it is released into the physical domain that is simultaneously being created by the erasure process itself.

The ontological reinterpretation is therefore as follows: *the erasure of a bit at the level of the informational substrate is the act by which informational potential is converted into physical actuality*. Before erasure, a bit in the statistical manifold is in a superposition of possible states — it contributes to the informational entropy of the substrate without corresponding to any definite physical configuration. The bit's existence at this stage is pre-

physical: it is a degree of freedom in the space of possibilities, not in the space of actualities. After erasure, the bit has been foreclosed — it no longer contributes to the informational entropy — and the energy $E = kT \ln 2$ has been released into the physical domain. This released energy is not merely a thermodynamic consequence; it *is* the physical actuality that the erasure brings into being.

"Landauer's Principle, in the standard reading, tells us how much forgetting costs. In the Theory of Entropicity, it tells us how much creation pays. The Landauer quantum is not the price of erasure; it is the yield of genesis."

— INFINITE Obidi, Theory of Entropicity (2026)

The transition effected by this reinterpretation is from *Landauer's Principle as a result about computation* to *Landauer's Principle as a law of physical ontogenesis*. It is analogous, in conceptual structure, to the transition from the mathematical description of curved surfaces (Gauss's differential geometry) to the physical claim that spacetime is itself curved (Einstein's general relativity). In both cases, an abstract mathematical structure — differential geometry in Einstein's case, information geometry in Obidi's — is identified not merely as a convenient description of physical phenomena but as the constitutive structure of physical reality itself.

This reinterpretation has immediate and profound consequences. It means that:

- The total mass-energy of the observable universe is the accumulated Landauer yield of all bit erasure events that have occurred since the beginning of entropic resolution.

- The formation of matter — particles, atoms, molecules, galaxies — corresponds to the clustering and condensation of Landauer quanta released by correlated erasure events.
- The geometry of spacetime is the large-scale pattern of the Fisher metric deformed by the distribution of bit erasure events across the statistical manifold.
- The arrow of time — the directionality of temporal experience — is the direction of ongoing entropic resolution in the informational substrate.

Each of these consequences is developed and formalized in the sections that follow.

4. The Core Framework of Obidi's Theory of Entropicity

With the mathematical foundations of information geometry (Section 2) and the ontological reinterpretation of Landauer's Principle (Section 3) in place, we are now prepared to present the core theoretical framework of Obidi's Theory of Entropicity in full detail. This section constitutes the central theoretical contribution of the monograph: it introduces and defines the key concepts of the theory, presents the emergence mechanism by which physical reality arises from the informational substrate, states the Entropicity field equations, derives the identification of mass with informational inertia, and presents the full hierarchy of emergent physical structure.

4.1 Defining Entropicity

The central concept that gives the theory its name is *Entropicity*. We define it here precisely and then unpack its implications in the sections that follow.

Definition (Entropicity): Entropicity is the intrinsic tendency of information-geometric configurations — points and distributions in the statistical manifold of the informational substrate — to undergo irreversible resolution: to transition from high-entropy, informationally superposed (undetermined) states to lower-entropy, actualized (determined) physical configurations, through the mechanism of irreversible bit erasure as specified by Landauer's Principle. Entropicity is a directed, generative tendency, not a passive statistical trend; it is the active force by which informational potential becomes physical actuality.

Several clarifications of this definition are necessary. First, Entropicity is not merely entropy production. Entropy production is a general feature of irreversible thermodynamic processes; Entropicity is the specific form of entropy production that occurs in the informational substrate and that, by the ontological reinterpretation of Landauer's Principle, generates physical reality. Second, Entropicity is irreversible by definition — the directionality of entropic resolution (from high-entropy informational superpositions to low-entropy actualized configurations) is what gives rise to the arrow of time. Third, Entropicity is a field phenomenon — it operates throughout the statistical manifold, not merely at localized points — and its dynamics are governed by the Entropicity field equations developed in Section 4.4.

Fourth, and most importantly for the physical interpretation: Entropicity is not merely a description of what happens; it is the *mechanism* by which what happens, happens. The Entropicity of the informational substrate is the reason that there is something rather than nothing — the reason that the

statistical manifold's potential configurations become actualized physical states rather than remaining forever in abstract superposition. In this sense, Entropicity plays in ToE the role that the Big Bang plays in standard cosmology: it is the primordial process, the generative act, the first cause (in a non-temporal sense) of physical reality.

4.2 The Informational Substrate: Structure and Properties

We now describe the structure and properties of the informational substrate — the information-geometric foundation of physical reality — in detail. The substrate possesses the following defining characteristics:

Statistical manifold structure. The informational substrate is a statistical manifold $M = \{p(x|\theta) : \theta \in \Theta\}$ of astronomical dimensionality. The dimension $\dim(\Theta)$ is of order 10^{122} , corresponding to the total number of informational degrees of freedom in the observable universe. Each point $\theta \in \Theta$ represents a complete informational state of the universe — a specification of the probability distributions over all possible physical configurations. Physical reality, as we observe it, is not a single point in M but a trajectory — an evolving specification of which configurations have been actualized — governed by the Entropicity field dynamics.

Fisher information metric. The substrate is equipped with the Fisher information metric $g_{ij}(\theta)$ as its natural Riemannian geometry. This metric encodes the statistical distinguishability of neighboring configurations and, as we shall show in Section 5, projects onto the metric tensor of emergent spacetime.

The Entropicity Field. The substrate is pervaded by a scalar field $\Phi_E : M \rightarrow \mathbb{R}$, which Obidi calls the *Entropicity Field*. This field encodes the local

tendency toward entropic resolution — the local "pressure" driving bit erasure events. High values of Φ_E correspond to regions of the manifold where configurations are under strong Entropicity pressure and erasure events are frequent; low values correspond to regions of relative informational stability. The dynamics of Φ_E are governed by the Entropicity field equation (Section 4.4).

Planck-scale granularity. The fundamental granularity of the informational substrate is the Planck scale. Each "bit cell" — the smallest irreducible unit of informational distinction — corresponds to a region of the statistical manifold of volume proportional to the Planck area $l_P^2 \approx 2.6 \times 10^{-70} \text{ m}^2$. Below this scale, the notion of distinct informational states — and hence the notion of spatial position — becomes undefined. This Planck-scale granularity is not imposed as an additional assumption; it follows from the requirement that the information density ρ_{info} be finite, which requires a minimum cell size in the statistical manifold.

Pre-geometric character. The substrate exists prior to and independently of the spacetime that emerges from it. The spatial and temporal coordinates of emergent spacetime are not coordinates on the statistical manifold; they are coarse-grained, collective variables that emerge in the limit of many correlated erasure events. The substrate has no preferred "location" in space (because space does not yet exist at the substrate level) and no preferred "moment" of time (because time is the direction of entropic resolution, which is itself a feature of the substrate dynamics rather than a pre-existing container).

Entropic coupling. The Fisher metric g_{ij} and the Entropicity field Φ_E are coupled by the entropic coupling constant λ , which has dimensions of (length)² and is related to the Planck area: $\lambda \sim l_P^2 / (k \ln 2)$. This coupling

encodes the rate at which informational pressure translates into Landauer energy release — the rate at which the Entropicity field drives physical genesis.

4.3 The Emergence Mechanism: Bit Erasure as Physical Genesis

We now present, in full and explicit detail, the five-step mechanism by which physical reality emerges from the informational substrate through Entropicity-driven bit erasure. This emergence mechanism is the operational core of ToE — the specific process by which the abstract ontological claims of the theory are realized.

Step 1 — The Initial High-Entropy Substrate

The informational substrate exists, at its initial (pre-physical) state, in a configuration of maximal entropy with respect to the Fisher metric. All informational degrees of freedom are in superposition — each bit cell of the statistical manifold encodes a probability distribution over all possible values, with no bit having a definite actualized value. This maximal-entropy state corresponds, in physical terms, to absolute nothingness: no particles, no forces, no spacetime geometry, no temporal direction. The substrate at this stage is a pure space of possibilities — an informational potential without physical actuality. In the language of quantum cosmology, this state resembles the Hartle-Hawking "no-boundary" state — a quantum superposition of all geometries with no preferred time direction — but at a level more fundamental than quantum geometry.

Step 2 — Entropicity Drives Irreversible Erasure

The Entropicity field Φ_E — which is non-zero throughout the substrate by virtue of the statistical manifold's inherent tendency toward configuration-space resolution — generates a pressure on the informational degrees of freedom. This pressure drives irreversible bit erasure events at the Planck scale: individual bit cells transition from superposed (high-entropy) states to definite (zero-entropy) values. The Entropicity field determines which bits are erased first, in what sequence, and with what spatial and temporal correlations. The field equation $\nabla_g^2 \Phi_E = \lambda \cdot \rho_{\text{info}}$ (developed in 4.4) governs the evolution of this pressure throughout the manifold. The initial perturbations that break the symmetry of the maximal-entropy state and trigger the first erasure events may be thought of as the informational analog of the quantum fluctuations that seed structure formation in standard cosmology.

Step 3 — Landauer Energy Release and the Physical Domain

Each bit erasure event releases a quantum of energy $E = kT_{\text{sub}} \ln 2$ into the physical domain, where T_{sub} is the effective temperature of the informational substrate — a quantity that ToE identifies with the Planck temperature $T_P = \sqrt{(\hbar c^5/Gk^2)} \approx 1.416 \times 10^{32}$ K at the cosmological onset of Entropicity. The Landauer energy release is not dissipated into a pre-existing thermal reservoir; rather, the act of release constitutes the physical domain. Each erasure event is a primordial creation event — an act of physical genesis at the Planck scale. The Landauer energy of the first erasure events was, in this picture, the primordial energy of the Big Bang — not a singular explosion but a distributed cascade of informational foreclosure events, each releasing a Landauer quantum into a nascent physical domain.

Step 4 — Entropic Condensation and the Formation of Matter

The Landauer quanta released by individual bit erasure events are not, in general, uniformly distributed across the emerging physical domain. The Entropicity field creates spatial and correlational structure in the erasure events: in regions of the statistical manifold where the Fisher metric has high curvature — where configurations are tightly constrained and statistically correlated — erasure events are clustered and correlated. These clusters of correlated erasure events produce localized concentrations of Landauer energy — concentrations that persist because the underlying informational configuration is stable (deeply entrenched in the statistical manifold). Obidi calls these stable, localized concentrations of Landauer energy *entropic condensation events*. They are identified with fundamental particles: each fundamental particle in the physical domain is the signature of an entropic condensation event — a stable, localized cluster of correlated bit erasures in the informational substrate. The different types of fundamental particles (quarks, leptons, gauge bosons) correspond to different topological and geometric structures of the condensation events in the statistical manifold.

Step 5 — Geometry of Erasure Patterns and the Emergence of Spacetime

As the Entropicity process proceeds and the informational substrate progressively resolves from high-entropy to lower-entropy configurations, the pattern of erasure events across the statistical manifold acquires a geometric structure. The Fisher metric of the statistical manifold, which tracks the statistical distinguishability of neighboring configurations, projects onto a large-scale geometric structure in the physical domain. Regions where erasure events are dense — where many bits per Planck area are being erased — correspond to regions of high information density and, by the field equation, high Entropicity field strength. These regions are the regions of high curvature in the emerging spacetime — the regions of strong

gravitational fields. Regions where erasure is sparse — where the informational substrate is relatively undisturbed — correspond to regions of low curvature: the empty space of the physical universe. In this way, the geometry of emergent spacetime IS the geometry of the erasure pattern in the statistical manifold, projected through the Fisher metric onto the physical domain.

4.4 The Entropicity Field Equations

The central dynamical equations of Obidi's Theory of Entropicity govern the evolution of the Entropicity field Φ_E across the statistical manifold M equipped with Fisher metric g_{ij} . The primary Entropicity field equation is:

$$\nabla_g^2 \Phi_E = \lambda \cdot \rho_{info} \text{ (Eq. 8) — Entropicity Field Equation (Primary)}$$

where ∇_g^2 is the Laplace-Beltrami operator with respect to the Fisher metric g_{ij} , defined by:

$$\nabla_g^2 \Phi_E = (1/\sqrt{g}) \partial_i (\sqrt{g} \cdot g^{ij} \partial_j \Phi_E) \text{ (Eq. 9) — Laplace-Beltrami Operator on Statistical Manifold}$$

with $g = \det(g_{ij})$; λ is the entropic coupling constant (dimensions of length², related to the Planck area: $\lambda \sim l_P^2 / k \ln 2$); and ρ_{info} is the local information density, defined as the magnitude of the entropy gradient:

$$\rho_{info} = |\nabla_g S_{info}| = \sqrt{g^{ij} \partial_i S_{info} \partial_j S_{info}} \text{ (Eq. 10) — Information Density}$$

where S_{info} is the information entropy of the local substrate configuration, measured in bits. Eq. (8) is formally analogous to the Poisson equation of Newtonian gravity ($\nabla^2 \phi = 4\pi G \rho$), with the Entropicity field Φ_E playing the role of the gravitational potential and the information density ρ_{info} playing the role

of the mass density. This analogy is not merely formal — it is the origin of the identification of gravity with an entropic gradient (Section 5.3).

The full (non-linear) Entropicity field equations, which govern the dynamics of Φ_E in the presence of back-reaction from the physical domain onto the substrate, are:

$$\nabla_g^2 \Phi_E - (1/c_E^2) \partial_t^2 \Phi_E = \lambda \cdot \rho_{info} - \mu \cdot \Phi_E^2 \text{ (Eq. 11) — Full Entropicity Field Equation}$$

where c_E is the propagation speed of Entropicity waves in the substrate (identified with the speed of light c in the physical domain), t is the entropic time coordinate (defined in Section 5.4), and μ is a self-coupling constant governing the non-linear back-reaction. The quadratic term $-\mu\Phi_E^2$ represents the saturation of Entropicity — the tendency of heavily erased regions (where the Entropicity field is large) to resist further erasure, corresponding physically to the stability of massive particles against spontaneous creation.

Solutions to the Entropicity field equations fall into two principal classes, corresponding to two distinct physical phenomena:

1. **Localized, stable solutions (solitons):** These are configurations in which Φ_E is concentrated in a small region of the statistical manifold and decays rapidly away from this region. These solutions are identified with fundamental particles — stable concentrations of Entropicity (and hence of Landauer energy) corresponding to the elementary excitations of physical reality.
2. **Long-range gradient solutions:** These are configurations in which Φ_E varies slowly over large regions of the statistical manifold, producing a long-range gradient. These solutions are identified with

the fundamental forces — in particular, gravity as the dominant long-range entropic gradient (Section 5.3).

The Entropicity field equations thus provide a unified description of both matter (solitonic solutions) and forces (gradient solutions) in terms of a single scalar field on the statistical manifold. This unification is one of the most striking features of the ToE framework: the distinction between matter and force is not fundamental but emergent, arising from the two qualitatively different types of solutions to the same field equation.

4.5 Mass as Informational Inertia

One of the most important and physically precise results of the Theory of Entropicity is the derivation of the concept of mass as a property of the informational substrate — specifically, as a measure of the resistance of an information-geometric configuration to further entropic resolution. We call this property *informational inertia*.

Consider an entropic condensation event — a stable, localized solution to the Entropicity field equations — corresponding to a fundamental particle. This configuration represents a region of the statistical manifold in which N bits have been irreversibly erased, producing a concentration of Landauer energy. The stability of the configuration implies that the remaining informational degrees of freedom in the region are in a highly ordered, low-entropy state — they have been "committed" to a definite physical configuration by the erasure events that produced the particle.

The resistance of this configuration to further entropic resolution — the difficulty of displacing or dissolving the condensation — is what we identify with the mass of the particle. A more massive particle corresponds to a deeper, more strongly entrenched informational configuration — one in

which more bits have been erased and the remaining degrees of freedom are more tightly correlated. The rest mass-energy mc^2 of the particle is therefore equal to the total accumulated Landauer energy of all the bit erasure events that constitute the particle:

$$mc^2 = N \cdot kT_{\text{sub}} \ln 2 \text{ (Eq. 12) — Mass-Energy as Landauer Sum (Informational Mass Formula)}$$

where N is the number of informational degrees of freedom foreclosed (bits erased) in the formation of the particle, and T_{sub} is the effective temperature of the informational substrate at the moment of the particle's formation. This is the *Informational Mass Formula* — one of the central results of ToE. It provides a purely informational derivation of mass-energy, reducing the concept of mass from an unexplained primitive of particle physics to a calculable property of the information-geometric structure of the particle's condensation event.

Eq. (12) has several important implications. First, it implies that the masses of fundamental particles are determined by the number of bits erased in their formation — a discrete, combinatorial quantity — which suggests that the spectrum of particle masses should be discrete, as indeed it appears to be. Second, it implies that massive particles are thermodynamically more stable than light particles, because the larger the number of erased bits, the deeper the entropic commitment of the configuration and the greater the Entropicity field strength needed to dissolve it. Third, it provides a natural explanation for the mass hierarchy problem of particle physics: the wide range of particle masses (from the near-zero neutrino mass to the heavy top quark mass) corresponds to a wide range of N values — from small, loosely bound condensation events to large, deeply committed ones.

The dynamical mass — the mass of a moving particle as seen from a reference frame in which the particle is in motion — receives a correction from the gradient of the Entropicity field along the particle's trajectory in the statistical manifold. This correction reproduces, in the appropriate limit, the relativistic mass formula $m(v) = m_0 / \sqrt{1 - v^2/c^2}$ of special relativity, providing a derivation of relativistic kinematics from the Entropicity framework (the detailed calculation is presented in Section 9).

4.6 The Hierarchy of Emergence

The Theory of Entropicity organizes the structure of physical reality into a hierarchy of levels, each level emerging from the one below through the progressive actualization of informational configurations. The hierarchy has six levels:

Level	Name	Description	Mathematical Structure
Level 0	Informational Substrate	The statistical manifold in its pre-physical, high-entropy state. No spacetime, no matter, no time — pure informational potential.	(M, g_{ij}) — Riemannian statistical manifold with Fisher metric
Level 1	Entropicity Field Dynamics	The Entropicity field Φ_E operating on the substrate, driving irreversible bit erasure events and releasing Landauer quanta. The	$\nabla^2_g \Phi_E = \lambda \cdot \rho_{\text{info}}$; Landauer energy $E = kT \ln 2$ per erasure

Level	Name	Description	Mathematical Structure
		primordial physical process.	
Level 2	Quantum Fields	Superpositions of partial-erasure configurations — regions of the substrate where some but not all bits have been erased. The quantum field is the physical correlate of the partially-resolved statistical manifold.	Quantum field operators as functions on the partially-resolved manifold; path integral over unresolved configurations
Level 3	Particles and Mass-Energy	Completed entropic condensation events — stable, localized concentrations of Landauer energy corresponding to fundamental particles. Mass-energy is the physical signature of informational foreclosure.	$mc^2 = N \cdot kT_{\text{sub}} \ln 2$; particle spectrum from solitonic solutions to Entropicity field equation
Level 4	Spacetime Geometry	The large-scale geometric structure emerging from the Fisher metric of the erasure pattern.	$g_{\mu\nu}(x) = \lim[\text{continuum}] g_{ij}(\theta)$; Riemannian geometry of emergent spacetime

Level	Name	Description	Mathematical Structure
		The spacetime metric tensor is the continuum limit of the Fisher metric projected onto the physical domain.	
Level 5	Classical Physics and General Relativity	The coarse-grained, macroscopic description of physical reality in the continuum limit of Level 4. The laws of classical mechanics, electrodynamics, and general relativity are theorems about the large-scale behavior of the Entropicity process.	Einstein field equations $G_{\mu\nu} = 8\pi G/c^4 \cdot T_{\mu\nu}$ as continuum limit; Newtonian mechanics as further limit

The hierarchy is not merely a classification scheme; it is an ontological ordering. Each level is constituted by the one below it — the higher level is literally made of the lower level, in the sense that its properties and laws are entirely determined by the dynamics of the lower level. The hierarchy is unidirectional: higher levels emerge from lower levels, but lower levels are not influenced by higher levels (modulo the back-reaction term in Eq. (11), which represents the small but non-negligible influence of macroscopic physical configurations on the substrate dynamics — the physical interpretation of quantum measurement, as discussed in Section 6.2).

5. Spacetime as an Emergent Information Geometry

The claim that spacetime is emergent — that it is not a fundamental container for physical events but a derived, coarse-grained description that arises from deeper structures — is one of the most radical and consequential propositions in contemporary theoretical physics. Obidi's Theory of Entropicity provides a precise and mathematically rigorous account of how spacetime emerges from the information-geometric substrate, how the metric tensor of general relativity arises from the Fisher metric of the statistical manifold, and how gravity — the curvature of spacetime — is an entropic gradient of the Entropicity field.

5.1 The Problem of Quantum Gravity and the Spacetime Emergence Paradigm

The central unsolved problem of theoretical physics is the reconciliation of general relativity and quantum mechanics. General relativity describes spacetime as a smooth, continuous, four-dimensional Lorentzian manifold with metric tensor $g_{\mu\nu}$, satisfying the Einstein field equations $G_{\mu\nu} = 8\pi G/c^4 \cdot T_{\mu\nu}$. The theory treats spacetime as a dynamical entity — curved by mass and energy, evolving according to deterministic equations — but as a fundamentally classical entity: it has no quantum superposition, no uncertainty principle, no discrete spectrum.

Quantum mechanics, by contrast, demands that physical systems exist in superpositions of states until measurement collapses the wavefunction to a definite outcome. It imposes the uncertainty principle $\Delta x \Delta p \geq \hbar/2$, which

prevents the simultaneous determination of position and momentum. It requires discrete spectra for bound systems. When one attempts to apply these quantum principles to spacetime itself — as one must if spacetime is physical and not merely a classical background — the result is a collection of devastating mathematical inconsistencies. The perturbative approach to quantum gravity, treating the graviton as a quantum field on a flat background spacetime, produces a non-renormalizable theory: divergences appear at every order of perturbation theory, and no finite number of counterterms can absorb them. The theory has no predictive power at high energies.

The deepest root of this incompatibility is that general relativity and quantum mechanics make contradictory ontological assumptions about spacetime. General relativity assumes spacetime is smooth and continuous, with a definite metric at every point. Quantum mechanics demands discreteness, superposition, and non-commutativity. These assumptions cannot both be correct if spacetime is fundamental. The resolution offered by Obidi's Theory of Entropicity is to deny the assumption that spacetime is fundamental: if spacetime is emergent — a coarse-grained, collective description that arises from the information-geometric substrate — then there is no conflict between the quantum behavior of matter and the classical behavior of spacetime. The quantum behavior of matter is the behavior of Level 2 (quantum fields — partially resolved informational configurations); the classical behavior of spacetime is the behavior of Level 4 (emergent geometry — the large-scale Fisher metric of resolved configurations). These are descriptions of the same underlying substrate at different levels of coarse-graining, and their apparent incompatibility is an artifact of treating them as fundamentally distinct.

The spacetime emergence paradigm — the view that spacetime arises from more primitive, non-spatiotemporal structures — has been advocated in

various forms by many researchers. Loop quantum gravity dissolves spacetime into a discrete network of spin-foam excitations. Causal set theory replaces the continuum of spacetime with a discrete partial order of events. String theory adds extra dimensions and replaces point particles with extended objects, but retains spacetime as a fundamental structure (modulo T-duality and holographic dualities). None of these approaches, however, provides a clear account of *why* spacetime emerges — what the generative mechanism is by which discrete, non-spatiotemporal structures give rise to the smooth, four-dimensional Lorentzian manifold of classical general relativity. ToE provides exactly this: the generative mechanism is Entropicity-driven bit erasure, and the emergent spacetime is the continuum limit of the Fisher metric of the erasure pattern.

5.2 From Fisher Metric to Spacetime Metric

The central mathematical claim of this section is: the metric tensor $g_{\mu\nu}(x)$ of emergent spacetime is, in the appropriate continuum limit and dimensional projection, the Fisher information metric $g_{ij}(\theta)$ of the underlying statistical manifold. The spacetime interval:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \text{ (Eq. 13) — Spacetime Interval (General Relativity)}$$

corresponds, under the emergence map $\varepsilon : M \rightarrow (\text{spacetime})$, to the Fisher information distance:

$$ds_F^2 = g_{ij}(\theta) d\theta^i d\theta^j \text{ (Eq. 14) — Fisher Distance (Statistical Manifold)}$$

between neighboring informational configurations θ and $\theta + d\theta$. The emergence map ε is not a simple coordinate transformation; it is a coarse-graining operation that averages over the microscopic (Planck-scale) details of the erasure pattern and projects the resulting large-scale information-

geometric structure onto a four-dimensional Lorentzian manifold. The four dimensions of spacetime — three spatial and one temporal — arise from the four dominant macroscopic modes of variation in the Fisher metric after coarse-graining. (Why exactly four? This is one of the open questions of ToE, discussed in Section 10.)

The physical interpretation of this correspondence is as follows. The Fisher distance ds^2_F between two neighboring configurations θ and $\theta + d\theta$ measures how statistically distinguishable they are — how much information separates them. In the physical domain, the spacetime interval ds^2 between two neighboring events x and $x + dx$ measures how separated they are in spacetime. These are the same thing, viewed from the substrate level and the physical level respectively: what we call "being at different places in space" or "being at different times" is, at the substrate level, "being at informationally different configurations."

Regions of high information density — where many bits are being erased per unit volume of the statistical manifold — correspond to regions where the Fisher metric is large, reflecting the high statistical distinguishability of neighboring configurations. In the physical projection, these regions correspond to regions of large spacetime curvature — strong gravitational fields. The Einstein field equations $G_{\mu\nu} = 8\pi G/c^4 \cdot T_{\mu\nu}$, in this picture, are the continuum-limit statement of the relationship between the information density ρ_{info} (encoded in $T_{\mu\nu}$) and the curvature of the emergent spacetime (encoded in $G_{\mu\nu}$). The derivation of Einstein's equations from the Entropicity field equations in the appropriate limit is presented in Section 9.

5.3 Gravity as Entropic Gradient: The ToE Derivation

In the Theory of Entropicity, gravity is not a fundamental force; it is the large-scale manifestation of the gradient of the Entropicity field across the statistical manifold. The gravitational force experienced by a body in the physical domain is the force that drives the body toward regions of lower Entropicity field strength — toward configurations of lower informational entropy, deeper in the potential well of the statistical manifold.

The gravitational force on a body of mass m in a region where the information entropy gradient is ∇S_{info} is:

$$F_{\text{grav}} = -T \cdot \nabla S_{\text{info}} \text{ (Eq. 15) — Gravitational Force as Entropic Gradient}$$

where T is the local substrate temperature (related to the Planck temperature and the local Entropicity field strength) and ∇S_{info} is the gradient of the information entropy of the local substrate configuration in the direction of the body's displacement. This expression is formally identical to the entropic force formula proposed by Verlinde (2011), but with a crucial difference: in ToE, the entropy S_{info} is identified explicitly with the entropy of the informational substrate configurations (rather than remaining a formal thermodynamic quantity), and the temperature T is the effective temperature of the Entropicity process (rather than an Unruh temperature or Hawking temperature).

In the non-relativistic limit, with the Entropicity field varying slowly and the information density distributed as a spherically symmetric mass M , Eq. (15) reproduces Newton's law of universal gravitation:

$$F_{\text{grav}} = -GMm / r^2 \text{ (Eq. 16) — Newton's Gravitational Law as ToE Limit}$$

The derivation proceeds by computing the information entropy gradient ∇S_{info} for a spherically symmetric mass concentration in the statistical manifold, using the Entropicity field equation (8) to relate ρ_{info} to the matter distribution, and then taking the non-relativistic, large-scale limit. Newton's gravitational constant G emerges as a combination of the entropic coupling constant λ , Boltzmann's constant k , and Planck's constant $\hbar$:

$$G = \lambda \cdot k \cdot \ln 2 / (2\pi \cdot \hbar \cdot l_P) \text{ (Eq. 17) — Gravitational Constant from ToE Parameters}$$

This derivation is significant because it demonstrates that the gravitational constant G is not a free parameter of the theory but a derived quantity — expressible in terms of the more fundamental parameters of the Entropicity framework. In principle, this provides a path toward predicting the value of G from first principles, though the precise computation requires knowledge of the entropic coupling constant λ that must itself be determined from observation (Section 10).

At the relativistic level — in the regime of strong gravitational fields and high information densities — the Entropicity field equation (11) predicts corrections to Eq. (15) that reproduce the full non-linear structure of Einstein's general relativity. The detailed derivation of the Einstein field equations from the Entropicity framework is given in Section 9. At extreme scales — near the Planck scale, or in the vicinity of black hole singularities — ToE predicts further corrections beyond general relativity (Section 8), reflecting the discrete, granular nature of the informational substrate.

5.4 Time as Entropic Direction

Among the deepest questions that any fundamental theory of physics must address is the nature of time. What is time? Why does it flow in one direction — from past to future — rather than the other? Why is the past fixed and the

future open? These questions are not adequately answered by any existing physical theory. General relativity treats time as a coordinate on an equal footing with space, with no preferred direction. Quantum mechanics has a time-symmetric Schrödinger equation, so the direction of time (the arrow) must be imposed as an additional assumption. Thermodynamics provides the most natural account — time flows in the direction of increasing entropy — but says nothing about why entropy should have been low in the past.

Obidi's Theory of Entropicity provides a complete, natural account of time and its directionality. In ToE, time is not a fundamental dimension — it is the direction of increasing entropic resolution in the informational substrate. The arrow of time is the arrow of Entropicity: the direction in which bit erasure proceeds. The past is the set of already-foreclosed configurations — the bits that have been erased, the possibilities that have been actualized. The future is the remaining space of unresolved configurations — the bits that have not yet been erased, the possibilities that remain open. The present is the wavefront of entropic resolution — the boundary between the already-actualized and the still-possible.

This account of time has several immediate virtues. It explains the asymmetry of time: time is directed not by any asymmetric fundamental law but by the irreversibility of bit erasure. Once a bit is erased, the information it encoded is gone — the foreclosure is permanent. This irreversibility is the physical basis of the arrow of time. The past is fixed because its bits have been erased and cannot be un-erased; the future is open because its bits have not yet been erased.

It also explains why entropy was low in the past. In ToE, the initial state of the informational substrate was one of maximal entropy (Step 1 of Section 4.3) — all bits unresolved, all configurations possible. But the *physical* entropy — the

entropy of the emergent physical world — was minimal at the beginning, because the emergent physical world was minimal: no particles, no structure, no matter. As Entropicity proceeds and bits are erased, the physical world grows richer (more particles, more structure, more organized matter), and the physical entropy grows. The low entropy of the early universe, in ToE, is not a mysterious initial condition to be explained by inflation or other mechanisms; it is the natural consequence of the fact that the physical world was young and few bits had been erased.

5.5 The Holographic Principle as a Theorem of ToE

The holographic principle — that the maximum information content of a region of space scales as the area of its boundary rather than its volume, in accordance with the Bekenstein-Hawking formula $S_{\text{BH}} = A / 4l_{\text{P}}^2$ — is one of the most profound results of modern theoretical physics. It was first proposed by 't Hooft in 1993 and developed by Susskind into the holographic principle, and it has received its most concrete realization in the AdS/CFT correspondence of string theory. In all existing treatments, the holographic principle is an independent result — a constraint on physical theories that must be satisfied, but whose deeper explanation remains unclear.

In Obidi's Theory of Entropicity, the holographic principle is not an independent postulate or an empirical constraint — it is a *theorem*, derivable from the structure of the Entropicity field and the information-geometric substrate. The derivation proceeds as follows.

The information content of a region R of the physical universe is, in ToE, the number of bits erased in the formation of that region — equivalently, the number of informational degrees of freedom in the statistical manifold that have been foreclosed in producing the physical configuration of R . By the Entropicity field equation (8), the information density ρ_{info} within R

determines the Entropicity field Φ_E via a Poisson-like equation. The boundary conditions of this equation — the values of Φ_E on the boundary ∂R of the region — determine the solution for Φ_E throughout the interior of R .

This is the key observation: the Entropicity field throughout the interior of R is completely determined by its values on the boundary ∂R . The physical information of the region is therefore encoded in the boundary conditions — in ∂R , not in the interior. The information content of R scales with the area of ∂R (the number of Planck-area cells on the boundary), not with the volume of R . Counting the Planck-area cells on ∂R gives:

$$S_{max}(R) = A(\partial R) / 4l_P^2 \text{ (Eq. 18) — Bekenstein-Hawking Entropy as ToE Theorem}$$

which is precisely the Bekenstein-Hawking formula. The factor of $1/4$ arises from the precise relationship between the Planck area and the entropy of a boundary bit cell in the Entropicity framework — a calculation given in detail in Section 9. The holographic principle thus emerges as a consequence of the Poisson-like structure of the Entropicity field equation and the Planck-scale granularity of the informational substrate.

6. Connecting ToE to Established Physics

A theoretical framework of the ambition and scope of Obidi's Theory of Entropicity must demonstrate its compatibility with — and, ideally, its derivation of — the established results of physics. This section performs exactly this demonstration: it shows how the Theory of Entropicity naturally encompasses, explains, and in several cases derives the central results of

thermodynamics, quantum mechanics, black hole physics, cosmology, and quantum information.

6.1 Thermodynamics: The Second Law as Entropicity Principle

The second law of thermodynamics — that the total entropy of an isolated system never decreases — is, in the standard treatments, an empirical generalization and a statistical theorem. It holds because there are overwhelmingly more high-entropy macrostates than low-entropy macrostates, so a system evolving randomly is overwhelmingly likely to be found in a higher-entropy state at later times. This statistical argument, while correct, does not explain *why* the universe started in a low-entropy state, nor does it explain why the second law applies to the universe as a whole.

In Obidi's Theory of Entropicity, the second law is not a statistical theorem about the likelihood of different macrostates; it is the expression of a fundamental ontological fact: bit erasure is irreversible. The second law of thermodynamics IS the Entropicity Principle: entropy increases in isolated systems because Entropicity is a one-way process. Bits, once erased, cannot be un-erased. Informational foreclosure is permanent. The direction of thermodynamic time is identical to the direction of Entropicity — the direction of bit erasure in the informational substrate.

This account of the second law is, in an important sense, more fundamental than the standard statistical account. It does not appeal to the overwhelming number of high-entropy macrostates (a counting argument that depends on the dimensionality of the phase space and the definition of macrostates). It appeals instead to the irreversibility of the physical process (bit erasure via Landauer's mechanism) that generates the physical world. The second law holds not because high-entropy states are statistically more accessible but

because the process of Entropicity, by which all physical states are generated, is inherently and irreversibly directed.

6.2 Quantum Mechanics: Superposition and Measurement as Partial Erasure

Quantum mechanics presents, at its heart, two profound conceptual puzzles: (i) the superposition principle — that quantum systems can exist in coherent superpositions of classically distinct states — and (ii) the measurement problem — that the act of measurement appears to "collapse" these superpositions to definite outcomes in a manner not described by the unitary Schrödinger equation. Both of these puzzles find natural and precise resolutions in Obidi's Theory of Entropicity.

Quantum superposition in ToE. A quantum system in a superposition of states $|\psi\rangle = \sum c_n |\psi_n\rangle$ corresponds, in the ToE framework, to a region of the informational substrate in which the relevant bits have not yet been erased — in which the informational degrees of freedom corresponding to the system's state remain unresolved. The wavefunction $\psi(x)$ is the physical correlate of the probability distribution $p(x|\theta)$ of the partially-resolved substrate configuration. The quantum amplitudes c_n are the square roots of the probabilities $p_n = |c_n|^2$ that each configuration will be actualized when the relevant bits are erased. Superposition is not a mysterious feature of reality; it is the physical signature of informational incompleteness — of bits that have not yet been erased.

Wavefunction collapse in ToE. Quantum measurement — the physical interaction that triggers wavefunction collapse — is, in ToE, a bit erasure event. When a quantum system interacts with a measuring apparatus (itself a physical system composed of many erased bits), the interaction constitutes an

irreversible correlation between the system and the apparatus. This correlation is an informational entanglement of the system's unresolved bits with the apparatus's already-resolved bits. The resolution of this entanglement — the actualization of one outcome — is an erasure event: one possibility is foreclosed, the others are discarded, and the Landauer energy of the erasure is dissipated into the apparatus's thermal degrees of freedom (which is why measurements generate heat and are irreversible). This is wavefunction collapse: a thermodynamic event, governed by Landauer's Principle, that converts informational potential into physical actuality.

This account of the measurement problem is one of the most valuable contributions of ToE. It resolves the measurement problem without any ad hoc modification of quantum mechanics (as in GRW collapse theories), without invoking many worlds (as in Everettian quantum mechanics), and without appealing to consciousness (as in some Copenhagen interpretations). Collapse is a physical, thermodynamic process — bit erasure — that occurs whenever a quantum system becomes irreversibly correlated with an environment large enough to constitute an effective heat bath. The decoherence program of Zurek and others is recovered as the classical limit of this process: decoherence is the regime in which erasure events occur too rapidly and densely for quantum coherence to be maintained.

6.3 The Bekenstein-Hawking Entropy and Black Holes

Black holes are the most extreme physical objects in the known universe — regions of spacetime where gravitational attraction is so strong that nothing, not even light, can escape. In general relativity, a black hole is characterized by its mass M , charge Q , and angular momentum J , and its event horizon is a surface of area $A = 16\pi G^2 M^2 / c^4$ (for a Schwarzschild black hole). The

Bekenstein-Hawking entropy $S_{\text{BH}} = A/4l_{\text{P}}^2 = kc^3A/4G\hbar$ is proportional to this area, and the Hawking temperature $T_{\text{H}} = \hbar c^3/8\pi GMk$ is inversely proportional to the black hole mass.

In Obidi's Theory of Entropicity, black holes are *maximal entropic condensation events* — regions where the Entropicity field has driven informational resolution to its maximum density, concentrating the maximum amount of Landauer energy in the minimum volume consistent with the Planck-scale granularity of the informational substrate. A black hole represents the limiting case of the emergence mechanism: a region in which the informational degrees of freedom have been so thoroughly erased — so deeply "committed" to a definite physical configuration — that no further information can be transmitted from the interior to the exterior. The event horizon is the boundary of this region of maximal informational commitment.

The Bekenstein-Hawking entropy $S_{\text{BH}} = A/4l_{\text{P}}^2$ counts, in the ToE framework, the number of unresolved bits at the horizon — the bits that are in the process of being erased by the Entropicity field but have not yet completed their erasure. These bits are suspended at the boundary between the maximally-erased interior (the black hole) and the partially-erased exterior (the surrounding spacetime). The factor of 1/4 in the formula arises from the geometry of the Entropicity field near the horizon: only one quarter of the Planck-area cells on the horizon surface are, at any given time, in the active phase of erasure (the others are either already erased or not yet reached by the Entropicity wavefront).

Hawking radiation, in ToE, is the slow Landauer leakage of horizon bits into the physical domain. As the Entropicity field gradually erases the horizon bits, their Landauer energy is released — not as a single burst but as a slow thermal radiation, with a temperature equal to the Hawking temperature T_{H} .

The information encoded in the Hawking radiation — the question of whether black hole evaporation is unitary — is resolved in ToE as follows: the information is never truly lost. It remains encoded in the geometry of the statistical manifold (in the pattern of which bits have been erased and how they were correlated) even after all the horizon bits have been erased and the black hole has evaporated. The physical information "in" the black hole is, at the substrate level, encoded in the informational configuration of the manifold — not in the physical interior of the black hole. Black hole evaporation is therefore a unitary process at the substrate level, even though it appears non-unitary at the level of the emergent physical description.

6.4 Dark Energy and the Cosmological Constant

Dark energy — the mysterious component of the universe's energy budget that drives the observed accelerating expansion — remains one of the deepest enigmas of modern cosmology. The standard cosmological model includes it as a cosmological constant Λ in the Einstein field equations, with a value of approximately $\Lambda \approx 1.1 \times 10^{-52} \text{ m}^{-2}$. The fine-tuning problem — why Λ is so small, and yet non-zero — is one of the most severe unsolved problems in theoretical physics, with the quantum field theory prediction for the vacuum energy density being larger than the observed value by some 120 orders of magnitude.

In Obidi's Theory of Entropicity, the cosmological constant Λ has a natural and simple explanation: it is the background Entropicity pressure of the informational substrate — the residual tendency of unresolved informational configurations to generate outward pressure on the emergent spacetime. As the Entropicity process proceeds and more bits are erased, the already-resolved physical domain expands (in the sense that the number of Landauer quanta in the physical domain increases). The still-unresolved configurations

of the substrate — the informational "dark" complement of the actualized physical world — exert an outward pressure on the boundary of the actualized domain. This pressure is what manifests in the physical domain as the cosmological constant.

The fine-tuning problem is resolved in ToE by observing that the value of Λ is not a free parameter of the theory but is determined by the current state of the Entropicity process — specifically, by the ratio of the number of erased bits (actualized matter and energy) to the number of unresolved bits (remaining informational potential). At early times, when few bits had been erased, this ratio was small and Λ was large. Over cosmic time, as more bits are erased and the ratio increases, Λ decreases. At the current cosmic epoch, the ratio takes a value that produces the observed small (but non-zero) Λ . The accelerating expansion of the universe is therefore the large-scale manifestation of the ongoing Entropicity process: the universe expands because the substrate continues to resolve, and the residual informational pressure of unresolved configurations pushes the spacetime geometry outward.

6.5 Quantum Entanglement as Shared Informational Configuration

Quantum entanglement — the phenomenon in which two particles, once having interacted, share a joint quantum state that cannot be factored into independent states of the individual particles — is often described as the most non-classical feature of quantum mechanics. Entangled particles exhibit correlations that violate Bell's inequalities, confirming that these correlations cannot be explained by any local hidden-variable theory. The non-locality of quantum entanglement — the fact that measuring one particle of an entangled pair instantly (or at least superluminally) affects the joint state —

has been one of the most debated features of quantum physics since Einstein, Podolsky, and Rosen raised the issue in 1935.

In Obidi's Theory of Entropicity, quantum entanglement finds a completely natural and non-mysterious explanation. Two entangled particles share a single informational configuration in the statistical manifold: their bits are correlated because they arose from the same erasure event or from a set of erasure events that were informationally linked — that is, the erasure of the bits corresponding to one particle was correlated (in the statistical manifold) with the erasure of the bits corresponding to the other particle.

The apparent non-locality of entanglement — the puzzle of how measuring one particle can affect the other regardless of the distance between them — dissolves completely in ToE. There is no "signal" transmitted between the particles; there is no superluminal causation. Rather, the two particles have never been independent informational configurations in the first place. At the level of the statistical manifold, they share a joint configuration — they are not two separate points in the manifold but two aspects of a single, correlated point. The physical distance between the particles in emergent spacetime is irrelevant at the substrate level, because spacetime is the emergent projection of the substrate and there is no spatial "distance" between configurations at the substrate level — only statistical distinguishability (Fisher distance). Two entangled particles are, at the substrate level, statistically close (or identical in the relevant degrees of freedom) regardless of how far apart they are in the physical domain.

7. Philosophical Implications

The Theory of Entropicity is not merely a physical theory; it is a philosophical revolution. Its implications extend far beyond the equations and predictions of theoretical physics into the deepest questions of metaphysics and the philosophy of science: What is the nature of physical existence? What are physical laws? Is the future genuinely open? What is the place of consciousness in the physical world? We address each of these questions in turn.

7.1 Information as the New Ontological Primitive

Western metaphysics has, since Aristotle, debated the question of what physical reality is ultimately made of. The ancient answers — fire, water, earth, air — gave way to Newtonian matter (extended, impenetrable atoms in absolute space and time), which gave way to Einsteinian fields and curved spacetime, which gave way to quantum fields and their excitations. At each stage, the ontological primitive — the most fundamental "stuff" of which everything is made — has become more abstract and further removed from ordinary intuition.

Obidi's Theory of Entropicity proposes the next step in this progression: the ontological primitive is information — specifically, informational configurations in a statistical manifold equipped with the Fisher metric. Physical reality, at its deepest level, is not made of particles, not of fields, not of spacetime, but of *probability distributions and their relations*. This is a form of structural realism — the view that what is fundamental is not material substance but abstract relational structure — taken to its logical conclusion.

This position should be carefully distinguished from idealism. The informational substrate of ToE is not mind-dependent; it does not require a

conscious observer; it exists (in the ontological sense appropriate to information-geometric structures) independently of whether any mind represents or observes it. The statistical manifold is an objective mathematical structure, not a construction of experience or thought. What ToE asserts is that this objective mathematical structure — and not any material substance — is the ultimate constituent of physical reality.

John Archibald Wheeler's "It from Bit" thesis is completed, but also corrected, by ToE. Wheeler was right that the "It" (physical reality) derives from the "Bit" (informational configurations). But he did not specify the mechanism. ToE specifies the mechanism: the "It" comes from the "Bit" through Entropicity-driven bit erasure, mediated by Landauer's Principle. The Bit does not merely *describe* the It; the Bit *generates* the It, through the ontological act of informational foreclosure.

7.2 The Nature of Physical Laws

In standard approaches to the philosophy of science, physical laws are typically treated as either brute facts — contingent regularities of an ultimately arbitrary universe — or necessary truths — but necessary in what sense, and derivable from what deeper principles? The laws of physics (Newton's laws, Maxwell's equations, Schrödinger's equation, Einstein's field equations) are extraordinarily precise, universal, and mathematically elegant. It has always seemed as though they *must* be derivable from something — but from what?

In Obidi's Theory of Entropicity, the answer is clear: physical laws are *theorems about the geometry of the statistical manifold and the dynamics of the Entropicity field*. They are not brute facts; they are mathematical consequences of the structure of the informational substrate. The laws of

physics hold not because the universe happens to be regular in these ways but because the information-geometric structure of the substrate demands it.

In particular, the conservation laws of physics — conservation of energy, conservation of momentum, conservation of angular momentum, conservation of charge — are, in ToE, expressions of the invariance symmetries of the Entropicity field dynamics. By Noether's theorem, applied to the action S_E of the Entropicity field (defined in Section 9), each continuous symmetry of the action generates a conserved current. The translational symmetry of the Fisher metric (invariance of the erasure dynamics under "shifts" in the statistical manifold corresponding to spatial translations in the physical domain) generates conservation of momentum. Time-translation symmetry (invariance under shifts in the entropic time direction) generates conservation of energy. Rotational symmetry generates conservation of angular momentum. The gauge symmetries of the Entropicity field generate conservation of charge. The conservation laws are thus not empirical regularities but mathematical necessities — consequences of the symmetry structure of the informational substrate.

7.3 Causality, Determinism, and the Open Future

ToE has a distinctive and important position on the questions of causality, determinism, and the openness of the future — questions that have been deeply entangled with the interpretation of quantum mechanics since Heisenberg's uncertainty principle first suggested that nature might be fundamentally indeterministic.

In ToE, the future is genuinely open — not merely unknown, but genuinely undetermined. This is because the future consists of informational configurations that have not yet been erased — possibilities that remain in superposition, unactualized. The Entropicity field equation determines the

probability distribution over which possibilities will be actualized (the higher the Entropicity field strength in a given region of the statistical manifold, the more likely that region's configurations will be erased first), but it does not determine *which specific configuration* will be erased at any given Planck-scale erasure event. The specific outcome of each erasure event has a probabilistic character governed by the probability distributions of the substrate configurations — and this probabilistic character is the origin of quantum indeterminism.

The universe, in ToE, is not a deterministic block — an eternally existing, four-dimensional spacetime in which all events (past, present, and future) exist simultaneously, with the experience of "flow" being an illusion. Rather, the universe is an *ongoing process of entropic resolution*, genuinely incomplete and genuinely open to the future. The four-dimensional spacetime of classical general relativity is the already-actualized portion of this process — the part of the statistical manifold that has been erased. The future is the unactualized remainder. The block universe view is wrong: the future is not "already there," waiting to be discovered; it is being created, moment by moment, by the ongoing Entropicity of the informational substrate.

Causality is preserved in ToE, but its character is subtler than in classical physics. Each bit erasure event has a cause — the local Entropicity field strength, which determines the probability of the event — but not a complete cause in the classical, deterministic sense. The causal structure of ToE is probabilistic at the Planck scale and deterministic (in the sense of definite field equations) at the macroscopic scale where many erasure events are averaged together.

7.4 Consciousness and Observation

One of the most persistent and contentious questions in the interpretation of quantum mechanics is whether conscious observation plays a special role in the collapse of the wavefunction. The Copenhagen interpretation's insistence that measurement outcomes are only defined relative to an observer, and Wigner's friend paradox, have led some to argue that consciousness is physically fundamental — that the physical world is constituted in part by the act of conscious observation.

ToE takes a clear and negative position on this question: conscious observation plays no special role in bit erasure or wavefunction collapse. Bit erasure occurs whenever a quantum system becomes irreversibly entangled with any environment large enough to constitute an effective heat bath — whether that environment is a conscious observer, an unconscious measuring apparatus, or a random collision with a gas molecule. The irreversibility of the erasure event — not the presence of conscious experience — is what triggers the actualization of a definite physical outcome.

Conscious experience is, in ToE, a high-level phenomenon: a particularly complex and integrated pattern of erasure events, corresponding to the information-processing activity of a brain. Consciousness is not more fundamental than matter; it is, if anything, less fundamental — it is a Level 5 (or even Level 6) emergent property of the physical world, which is itself a Level 3–5 emergent property of the informational substrate. This position resolves the measurement problem without invoking consciousness, preserves the objective, observer-independent character of physical reality, and avoids the incoherence of making physics dependent on an ill-defined concept such as "conscious observation."

8. Predictions and Empirical Consequences

Any theoretical framework of the ambition of Obidi's Theory of Entropicity must, in order to qualify as a scientific theory rather than merely a philosophical speculation, yield specific, falsifiable empirical predictions that distinguish it from competing theories. This section presents five families of empirical predictions of ToE, organized by the physical domain in which they are expected to manifest. These predictions are not post-hoc accommodations of known results; they are genuine predictions that, in several cases, differ from the predictions of standard general relativity and quantum field theory in ways that are in principle testable with current or near-future observational technologies.

8.1 Deviations from Standard Thermodynamics at Planck Scale

The informational substrate of ToE is granular at the Planck scale — individual bit cells of size l_P^2 are the fundamental units of informational structure. This granularity must become apparent in the thermodynamic properties of physical systems at energies approaching the Planck energy $E_P = \hbar c / l_P \approx 1.956 \times 10^9 \text{ J} \approx 1.22 \times 10^{19} \text{ GeV}$. At such energies, the continuum approximation that underlies standard thermodynamics breaks down, and corrections from the discrete structure of the informational substrate appear.

Specifically, ToE predicts that the heat capacity of Planck-scale systems — systems whose thermal energy is comparable to the Planck energy — deviates from the classical equipartition prediction by corrections of order $(E/E_{\text{Planck}})^2$. The modified heat capacity for a system of energy $E \ll E_{\text{Planck}}$ takes the form:

$$C(E) = C_{\text{classical}} \cdot [1 - \beta \cdot (E/E_{\text{Planck}})^2 + O((E/E_{\text{Planck}})^4)] \quad (\text{Eq. 19}) - \text{Modified Heat Capacity at Planck Scale}$$

where β is a dimensionless coefficient of order unity determined by the geometry of the statistical manifold near the Planck scale. For $E \ll E_{\text{Planck}}$, the correction is negligible, recovering standard thermodynamics. As $E \rightarrow E_{\text{Planck}}$, the correction becomes significant and $C(E)$ deviates measurably from the classical prediction. The sign of the correction (a decrease in heat capacity relative to classical predictions) reflects the saturation of Entropicity — the resistance of maximally-erased configurations to further erasure.

While direct measurement at the Planck energy scale is far beyond current experimental capabilities, indirect evidence for this correction may be sought in the thermodynamics of black holes near the Planck mass ($M_{\text{Planck}} \approx 2.18 \times 10^{-8}$ kg), in the final stages of black hole evaporation, where Hawking temperature approaches the Planck temperature and the Planck-scale granularity of the informational substrate becomes directly relevant.

8.2 Modified Dispersion Relations

Because spacetime is emergent from the information-geometric substrate and the substrate has a fundamental Planck-scale granularity, the energy-momentum dispersion relation for photons and other elementary particles is predicted to receive small corrections at high energies. In standard special relativity, the dispersion relation for massless particles (photons) is exact: $E^2 = p^2 c^2$. In ToE, the discrete structure of the informational substrate introduces corrections at energies approaching the Planck scale:

$$E^2 = p^2 c^2 + \alpha \cdot (E^3/E_{\text{Planck}}) + \gamma \cdot (E^4/E_{\text{Planck}}^2) + \dots \quad (\text{Eq. 20}) - \text{Modified Dispersion Relation (ToE Prediction)}$$

where α and γ are dimensionless parameters of order unity encoding the first- and second-order corrections from the Planck-scale structure of the substrate. The coefficient α is particularly important: if $\alpha \neq 0$, then high-energy photons travel at slightly different speeds than low-energy photons — a phenomenon called energy-dependent photon velocity or Lorentz invariance violation at the Planck scale.

This prediction is testable via gamma-ray burst (GRB) observations. Gamma-ray bursts are cosmological events at distances of billions of light-years that emit photons across a wide range of energies simultaneously. If high-energy photons travel slightly faster or slower than low-energy photons (by an amount proportional to their energy, as predicted by Eq. (20)), the difference in arrival time over cosmological distances becomes measurable. The Fermi Gamma-ray Space Telescope has already placed constraints on α at the level $|\alpha| \lesssim O(1)$ from GRB 090510 and other events, limits that are just beginning to reach the theoretical prediction range of first-order Planck-scale effects. ToE predicts a specific sign and magnitude for α (determined by the entropic coupling constant λ) that can be tested against these observations. Future gamma-ray observatories with greater sensitivity and broader energy coverage will provide definitive tests.

8.3 Minimum Length and Minimum Time

The discrete, granular nature of the informational substrate implies the existence of a minimum meaningful length and a minimum meaningful time. Below the Planck length $l_P \approx 1.616 \times 10^{-35}$ m, the concept of spatial position becomes undefined in ToE — there are no resolved informational configurations, and hence no emergent spatial structure, at sub-Planck scales. Similarly, below the Planck time $t_P \approx 5.391 \times 10^{-44}$ s, the concept of temporal interval becomes undefined.

These minimum scales are predictions of ToE, but they are also predictions of many other approaches to quantum gravity (loop quantum gravity, string theory, doubly special relativity). What distinguishes ToE's prediction is the specific relationship between the minimum length, the entropic coupling constant λ , and the Landauer energy: $l_p = \sqrt{(\lambda \cdot k \ln 2 / \hbar)}$. This relationship provides a specific value for λ (once l_p is taken from measurement), and this value can be checked for consistency with the other predictions of ToE (such as the gravitational constant derivation in Eq. (17)).

The minimum length also implies a generalized uncertainty principle (GUP) for position and momentum measurements:

$$\Delta x \Delta p \geq \hbar/2 \cdot [1 + \delta \cdot (\Delta p / p_{\text{Planck}})^2] \text{ (Eq. 21) — Generalized Uncertainty Principle (ToE)}$$

where $p_{\text{Planck}} = \hbar/l_p$ is the Planck momentum and δ is a dimensionless constant of order unity derived from the geometry of the statistical manifold. The standard Heisenberg uncertainty principle is recovered for $\Delta p \ll p_{\text{Planck}}$. The GUP correction becomes relevant at Planck-scale momenta and has potential consequences for very high-energy scattering experiments and the thermodynamics of black holes.

8.4 Information Content of the Observable Universe

The Theory of Entropicity predicts a specific value for the total information content of the observable universe — the total number of bits that have been erased (actualized) since the onset of Entropicity. This prediction follows from applying the Bekenstein bound to the observable universe as a whole.

The observable universe has a Hubble radius of approximately $R_H \approx 4.4 \times 10^{26}$ m. Its total Bekenstein entropy (maximum information content) is:

$$S_{max} = \pi R_H^2 / l_P^2 \approx 10^{122} \text{ bits (Eq. 22) — Total Information Content of the Observable Universe}$$

This estimate of approximately 10^{122} bits is consistent with the observed properties of the universe: the number of particles in the observable universe (approximately 10^{80}) multiplied by the number of bits needed to specify the quantum state of each particle (of order 10^{40} bits per particle, accounting for all quantum numbers and spatial degrees of freedom) gives a total of approximately 10^{120} – 10^{122} bits, consistent with the Bekenstein bound prediction. The precise agreement between these two independent estimates — the geometric Bekenstein bound and the particle-counting estimate — is, in ToE, not a coincidence but a consequence of the holographic structure of the Entropicity process: the number of bits erased is bounded by the area of the emerging spacetime, which is the holographic bound.

8.5 Entropic Corrections to Gravitational Dynamics

Perhaps the most observationally accessible prediction of ToE is the existence of small, measurable corrections to Newtonian gravity and general relativity at galactic scales, arising from the background Entropicity pressure of the informational substrate. The Entropicity field Φ_E has a non-zero background value throughout the statistical manifold — the residual Entropicity pressure of the unresolved informational configurations (which we identified in Section 6.4 with the cosmological constant). This background value produces a slight modification of the gravitational field at large distances from a mass concentration.

Specifically, the gravitational acceleration predicted by ToE in the low-acceleration regime (where standard Newtonian gravity gives accelerations much smaller than a characteristic acceleration scale $a_E = c \cdot \lambda^{-1/2} \approx 10^{-10}$ m/s²) is:

$$a_{\text{ToE}}(r) = a_{\text{Newton}}(r) \cdot [1 + (a_{\text{E}}/a_{\text{Newton}})^{1/2}]^{1/2} \text{ (Eq. 23) — ToE Modified Gravitational Acceleration}$$

In the regime $a_{\text{Newton}} \gg a_{\text{E}}$, Eq. (23) reduces to standard Newtonian gravity. In the regime $a_{\text{Newton}} \ll a_{\text{E}}$, the ToE prediction deviates significantly from Newtonian gravity, approaching $a_{\text{ToE}} \rightarrow \sqrt{a_{\text{Newton}} \cdot a_{\text{E}}}$. This low-acceleration regime corresponds precisely to the outer regions of galaxies, where the anomalous rotation curves are observed — where galactic rotation speeds remain flat at large radii rather than falling off as Kepler's law predicts for a finite mass distribution.

The prediction of Eq. (23) is strikingly similar to the empirical formula of Modified Newtonian Dynamics (MOND), proposed by Milgrom in 1983. MOND was introduced as a purely phenomenological fit to galactic rotation curves; ToE provides it with a theoretical foundation as a consequence of the background Entropicity pressure. This represents a major potential success of the ToE framework: it provides an information-theoretic, thermodynamic explanation for the galactic-scale gravitational anomalies that are standardly attributed to dark matter, without requiring any unseen mass. The critical observational tests — including measurements of galactic rotation curves at different redshifts, the galaxy-cluster dynamics, and gravitational lensing — will discriminate between the dark matter hypothesis and the ToE/Entropicity correction prediction.

9. Mathematical Appendix

This appendix collects the key mathematical results of the Theory of Entropicity in a self-contained, formally rigorous presentation. All claims

made in the body of the monograph that rely on specific mathematical results are supported here.

A1. Formal Definition of the Statistical Manifold and Fisher Metric

Definition A1 (Statistical Manifold).

Let X be a measurable sample space and let $\Theta \subseteq \mathbb{R}^n$ be an open subset of Euclidean n -space. Define the statistical manifold M as the set of probability density functions:

$$M = \{ p(x|\theta) : \theta \in \Theta, p(\cdot|\theta) >$$

$$0 \text{ a.e.}, \int p(x|\theta) dx = 1 \}$$

M is endowed with the differentiable structure of Θ via the parameterization $\theta \mapsto p(\cdot|\theta)$, assumed to be a smooth injective map. The Fisher information matrix at $\theta \in \Theta$ is the $n \times n$ matrix with components:

g

$_{ij}$

$$(\theta) = \int$$

X

$$p(x|\theta) [\partial$$

$_i$

$$\log p(x|\theta)][\partial$$

$_j$

$$\log p(x|\theta)] dx$$

Theorem A1 (Positive Definiteness). Under mild regularity conditions (differentiability and integrability of the score functions $\partial_i \log p$), $g_{ij}(\theta)$ is positive semi-definite at every $\theta \in \Theta$, and positive definite if the distributions $p(\cdot|\theta)$ are identifiable (distinct parameter values give distinct distributions). Under these conditions, (M, g) is a Riemannian manifold.

Proof sketch. For any vector $v \in \mathbb{R}^n$, $v^i g_{ij} v^j = \int p(x|\theta) [v^i \partial_i \log p]^2 dx = \mathbb{E}[(v \cdot \nabla_\theta \log p)^2] \geq 0$, with equality iff $v \cdot \nabla_\theta \log p = 0$ a.s., which (under identifiability) holds only for $v = 0$. $\square$

A2. The Cramér-Rao Bound as a Geometric Theorem

The Cramér-Rao bound is one of the fundamental results of statistical inference. It states that no unbiased estimator of the parameter θ can achieve a variance smaller than the inverse of the Fisher information. In the geometric language of statistical manifolds, this is a statement about the minimum curvature of the statistical manifold consistent with efficient estimation.

Theorem A2 (Cramér-Rao Bound). Let $T(x)$ be any unbiased estimator of θ^i (so that $\mathbb{E}[T(x)] = \theta^i$). Then:

$$\text{Var}(T) \geq [g^{-1}(\theta)]_{ii} \text{ (Eq. A1) — Cramér-Rao Bound}$$

where $[g^{-1}]_{ii}$ is the i -th diagonal element of the inverse Fisher metric matrix. Geometrically: the variance of any unbiased estimator of θ^i is bounded below by the length (in the Fisher metric) of the coordinate vector $\partial/\partial\theta^i$. Efficient

estimators — those achieving the bound — exist if and only if the statistical manifold is dually flat in a neighborhood of θ . This is the geometric significance of the bound: achievability is equivalent to dual flatness.

A3. Derivation of the Entropicity Field Equation from a Variational Principle

The Entropicity field equation (Eq. 8) is derived from the following action functional defined on the statistical manifold (M, g) :

$$S_E[\Phi_E] = \int_M \left[\frac{1}{2} g^{ij} \partial_i \Phi_E \partial_j \Phi_E + V(\Phi_E, \rho_{info}) \right] dVol_g \text{ (Eq. A2) — Entropicity Action Functional}$$

where $dVol_g = \sqrt{g} d^n \theta$ is the Riemannian volume element with respect to the Fisher metric, and $V(\Phi_E, \rho_{info})$ is the potential:

$$V(\Phi_E, \rho_{info}) = -\lambda \cdot \rho_{info} \cdot \Phi_E + \frac{1}{2} \mu \cdot \Phi_E^2 \text{ (Eq. A3) — Entropicity Potential}$$

Taking the functional derivative of S_E with respect to Φ_E and setting it to zero (the Euler-Lagrange equation):

$$\delta S_E / \delta \Phi_E = -\nabla_g^2 \Phi_E - \lambda \rho_{info} + \mu \Phi_E = 0 \text{ (Eq. A4) — Euler-Lagrange Equation for Entropicity Field}$$

In the linear regime ($\mu \Phi_E \ll \lambda \rho_{info}$), this reduces to the primary Entropicity field equation:

$$\nabla_g^2 \Phi_E = \lambda \rho_{info} \text{ (Eq. A5) — Primary Entropicity Field Equation (reproduced)}$$

The full non-linear equation (Eq. 11) is recovered when the time-derivative term (from the kinetic term in the action) and the self-coupling term (from the $\mu \Phi_E^2$ term in the potential) are retained.

A4. Reduction to Einstein's Field Equations in the Continuum Limit

In the continuum, low-information-density limit — when the number of erasure events per Planck volume is large and the Fisher metric varies slowly on scales much larger than l_P — the Entropicity field equation reduces to Einstein's field equations. The reduction proceeds in three steps:

3. **Projection:** The high-dimensional Fisher metric $g_{ij}(\theta)$ is projected, by the coarse-graining map ε , onto a four-dimensional Lorentzian metric $g_{\mu\nu}(x)$, where the four macroscopic coordinates x^μ are the dominant collective modes of variation of the informational configuration.
4. **Curvature identification:** The Ricci tensor $R_{\mu\nu}$ of the projected metric $g_{\mu\nu}$ is identified with the second-order variation of the Entropicity field Φ_E in the corresponding macroscopic directions: $R_{\mu\nu} = -(\partial^2 \Phi_E / \partial x^\mu \partial x^\nu)$ (in normal coordinates at a point).
5. **Source identification:** The information density ρ_{info} is identified with the trace of the energy-momentum tensor $T_{\mu\nu}$ via: $\rho_{\text{info}} = (8\pi G/c^4) \cdot T$, where $T = g^{\mu\nu} T_{\mu\nu}$.

Under these identifications, the Entropicity field equation $\nabla^2_g \Phi_E = \lambda \rho_{\text{info}}$ reduces, after appropriate rescaling and the addition of the cosmological term, to the trace of the Einstein field equations:

$$R - 2\Lambda = -(8\pi G/c^4) \cdot T \text{ (Eq. A6) — Trace of Einstein's Field Equations}$$

The full tensorial Einstein equation $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ is recovered by performing the reduction at the level of the full Entropicity action functional (Eq. A2), projecting the action onto the four-dimensional space, and identifying the resulting projected action with the Einstein-Hilbert action $\int (R - 2\Lambda) \sqrt{|g|} d^4x / (16\pi G/c^4)$. The cosmological constant Λ arises from the

background value of the Entropicity potential $V(\Phi_E, 0)$ in the vacuum — the residual Entropicity pressure of unresolved configurations.

A5. Derivation of the Bekenstein-Hawking Entropy Formula from Entropicity Field Boundary Conditions

Consider a spherical region R of the physical universe with boundary ∂R of area A . The Entropicity field Φ_E satisfies the field equation (A5) in the interior of R with boundary condition $\Phi_E|_{\partial R} = \Phi_{\text{boundary}}$. By the uniqueness of solutions to elliptic equations, the interior solution is completely determined by the boundary values. Counting the independent boundary degrees of freedom: the boundary ∂R is a two-dimensional surface discretized at the Planck scale, with each Planck-area cell (of area l_p^2) supporting one independent bit of boundary information. The total number of boundary bits is:

$$N_{\text{boundary}} = A / l_p^2 = A / (G\hbar/c^3) \text{ (Eq. A7) — Number of Boundary Bits}$$

Each boundary bit contributes $k \ln 2$ to the entropy (by Landauer's Principle). The maximum entropy is therefore:

$$S_{\text{max}} = k \ln 2 \cdot N_{\text{boundary}} = k \ln 2 \cdot A / l_p^2 \text{ (Eq. A8)}$$

The standard Bekenstein-Hawking formula $S_{\text{BH}} = kc^3A/(4G\hbar) = kA/(4l_p^2)$ differs from Eq. (A8) by the factor $1/(4 \ln 2)$. This factor is accounted for by the observation that each Planck-area boundary cell does not support a pure bit (two-state system) but a qubit in a thermal state at the Hawking temperature, with an effective entropy per cell of $k/(4 \ln 2)$ rather than $k \ln 2$. The thermal factor $1/(4 \ln^2 2)$ is derived from the thermal statistics of the boundary bits in the Entropicity framework — a calculation that involves the partition function

of the Entropicity field at the horizon temperature and that reproduces the exact Bekenstein-Hawking coefficient 1/4.

A6. Derivation of the Informational Mass Formula

The rest mass-energy of a particle is, in ToE, the total accumulated Landauer energy of the bit erasure events constituting the particle's entropic condensation event. Let N be the number of bits erased in forming a particle, and let T_{sub} be the effective substrate temperature at the moment of formation. Each bit erasure releases energy $E = kT_{\text{sub}} \ln 2$. The total rest energy is:

$$mc^2 = N \cdot kT_{\text{sub}} \ln 2 \text{ (Eq. A9) — Informational Mass Formula}$$

The number N of erased bits for a particle of mass m is:

$$N = mc^2 / (kT_{\text{sub}} \ln 2) = mc^2 / E_{\text{Landauer}} \text{ (Eq. A10)}$$

For a proton ($m = 1.673 \times 10^{-27}$ kg), taking $T_{\text{sub}} = T_{\text{Planck}} \approx 1.416 \times 10^{32}$ K, $E_{\text{Landauer}} = kT_{\text{P}} \ln 2 \approx 1.35 \times 10^9$ J, and $mc^2 = 1.503 \times 10^{-10}$ J, one finds $N \approx 1.1 \times 10^{-19}$. This small number reflects the fact that a proton's mass is a tiny fraction of the Planck energy — only a very small number of equivalent Planck-temperature bit erasures are needed to constitute a proton's mass. For macroscopic objects, N becomes astronomically large. The discrete, integer character of N provides a rationale for the discreteness of the mass spectrum of fundamental particles — though the precise mechanism by which specific integer values are selected is a topic for future theoretical development.

10. Discussion: Open Questions and Future Directions

The Theory of Entropicity, as presented in this monograph, constitutes a complete first statement of the theoretical framework — a full articulation of the central ideas, the mathematical structure, the connections to established physics, the philosophical implications, and the empirical predictions. Yet any honest theoretical monograph must acknowledge the open questions that remain — the lacunae in the framework where further work is required and the frontiers where the theory's future development lies. This section addresses the most important of these.

The quantum emergence problem. The Theory of Entropicity provides a natural account of how classical spacetime (Level 4) and classical physics (Level 5) emerge from the informational substrate through the Entropicity process. However, the precise mechanism by which the statistical manifold gives rise to the specific mathematical structure of quantum fields (Level 2) — with their Hilbert space formalism, operator algebra, and gauge symmetries — remains to be worked out in full. The identification of quantum superpositions with unresolved informational configurations (Section 6.2) is conceptually compelling, but the precise mathematical correspondence between the functional integral over unresolved configurations and the standard path integral formulation of quantum field theory requires further development. In particular, the question of how the specific gauge groups ($U(1)$, $SU(2)$, $SU(3)$) of the Standard Model arise from the geometry of the statistical manifold is an important open problem.

Pre-cosmological information theory. Obidi's Theory of Entropicity describes the informational substrate as existing "prior to" spacetime in an

ontological sense, in its initial high-entropy, pre-physical state. But what determined the initial state of the substrate? What are the "initial conditions" of the statistical manifold before Entropicity begins? And what — if anything — determined the specific structure of the manifold (its dimension, its topology, its initial information density) that gives rise to the specific physical universe we observe? These are questions of pre-cosmological information theory — the study of the structure of the informational substrate prior to the onset of Entropicity. This is perhaps the deepest open question in the ToE program and may require entirely new mathematical and conceptual tools to address.

Relationship to other approaches. How does ToE relate to the major existing programs for quantum gravity and foundational physics? *Loop quantum gravity* (Rovelli, Smolin, Thiemann) dissolves spacetime into discrete spin-foam networks — a picture that has formal similarities to the Planck-scale bit cells of the ToE substrate, but is derived from the quantization of general relativity rather than from information geometry. The relationship between spin-foam amplitudes and Entropicity field dynamics is a potentially fruitful area for future research. *Causal set theory* (Bombelli, Lee, Myrheim, Sorkin) replaces the spacetime continuum with a discrete partial order of events — the causal set. The causal structure of a causal set has informational overtones (each causal link represents a unit of causal information), and the connection between causal sets and the ordered structure of bit erasure events in ToE deserves investigation. *String theory* provides the AdS/CFT correspondence — a concrete realization of the holographic principle — and the holographic entanglement entropy formula (Ryu-Takayanagi formula), which resonates strongly with the ToE derivation of the Bekenstein-Hawking entropy. Whether string theory can be

reinterpreted as a specific implementation of ToE at the level of the quantum emergence problem is an intriguing possibility.

The entropic coupling constant. The entropic coupling constant λ appears throughout the ToE framework as a fundamental parameter relating the information density ρ_{info} to the Entropicity field strength Φ_E . In the present formulation, λ is related to the Planck area and Boltzmann's constant ($\lambda \sim l_P^2 / k \ln 2$), but this relationship is a dimensional estimate rather than a derivation. Whether λ can be derived from first principles — perhaps from the geometry of the statistical manifold itself, through a self-consistency condition on the Entropicity field equations — or whether it must remain a measured fundamental constant is an important open question. If λ is derivable, the ToE framework would achieve a degree of mathematical self-sufficiency — all physical constants determined by the geometry of information — that would represent a remarkable theoretical achievement.

Direct observation of Planck-scale informational structure. The most fundamental predictions of ToE — the granularity of the informational substrate at the Planck scale, the existence of individual bit erasure events as primordial physical acts — are far beyond the reach of current experimental technology. The Planck energy (10^{19} GeV) is some fifteen orders of magnitude above the highest energies reached by current particle accelerators. However, as discussed in Section 8, the indirect signatures of Planck-scale physics — modified dispersion relations, Planck-scale corrections to black hole thermodynamics, entropic corrections to galactic dynamics — are accessible with current or near-future observational instruments. The development of more sensitive gamma-ray observatories, gravitational wave detectors, and precision tests of gravitational dynamics at galactic scales will provide progressively stronger constraints on and tests of the ToE

predictions, building an empirical case for or against the framework over the coming decades.

11. Conclusion

This monograph has presented, in full theoretical detail, Obidi's Theory of Entropicity — a framework proposing that information geometry is not merely a description of physical reality but its constitutive foundation, and that the irreversible erasure of information at the Planck-scale informational substrate is the primordial act from which the universe's entire material and geometric structure arises. Let us draw the principal threads of the argument together.

The central thesis, stated at the outset and developed across eleven sections of theoretical construction, may now be appreciated in its full depth. Physical reality — the matter, energy, forces, and spacetime that constitute the observable universe — is not primary. It is emergent. It arises, through a precise and mathematically characterized process, from a deeper level of reality: a statistical manifold of astronomical dimensionality, equipped with the Fisher information metric, in which all possible configurations of physical reality are encoded as probability distributions. This informational substrate is pre-geometric — it does not exist in spacetime; spacetime exists in it, as an emergent projection of its geometry.

The generative mechanism — the process by which informational potential becomes physical actuality — is Entropicity: the irreversible, directional tendency of the informational substrate to undergo bit erasure, in accordance with Landauer's Principle elevated from computational thermodynamics to

ontological law. Each bit erased at the substrate level releases a quantum of Landauer energy — $kT \ln 2$ — into the physical domain. Matter is the condensation of correlated Landauer quanta; mass is the informational inertia of entropic condensation events; force is the gradient of the Entropicity field across the statistical manifold; spacetime geometry is the Fisher metric of the erasure pattern in the continuum limit; time is the direction of entropic resolution; and gravity is the entropic gradient that drives systems toward regions of lower informational entropy.

The hierarchy of emergence — from the informational substrate (Level 0) through Entropicity field dynamics (Level 1), quantum fields (Level 2), particles and mass-energy (Level 3), spacetime geometry (Level 4), to classical physics and general relativity (Level 5) — provides a complete organizational account of the known physical world, in which each level is rigorously constituted by the dynamics of the level below it. The major results of established physics — the second law of thermodynamics, quantum superposition and measurement, the Bekenstein-Hawking black hole entropy, the holographic principle, the cosmological constant, and quantum entanglement — all find natural, unified explanations within this framework.

The pivotal role of Landauer's Principle cannot be overstated. It is the bridge between the abstract and the concrete — the principle that specifies exactly how much physical reality (how much energy) each act of informational foreclosure creates. Without Landauer's Principle, the identification of information with physical reality would remain a metaphor. With it, the identification becomes a quantitative, mechanistically precise claim: information becomes matter at a rate of $kT \ln 2$ per bit erased. The universe's mass-energy budget is the Landauer yield of its entire history of entropic resolution.

What does it mean, for science and for human understanding, if physical reality is emergent from information geometry in the manner proposed by the Theory of Entropicity? It means, first, that the long quest for a unified theory of physics — a theory reconciling quantum mechanics, thermodynamics, and general relativity — finds its answer not in a more powerful quantum field theory or a more elaborate theory of strings and extra dimensions, but in a radical reconceptualization of the ontological status of spacetime itself. Spacetime is not the arena of physics; it is a product of physics — the coarse-grained residue of an informational process that operates at a level more fundamental than any spatial or temporal description.

It means, second, that the physical universe is not a deterministic machine grinding forward along a predetermined trajectory, nor a random collection of events with no intrinsic order. It is an *ongoing creative process* — a process of informational genesis in which the future is genuinely open, each moment of time is the actualization of one possibility among many, and the structure of physical reality is being built, bit by bit, through the irreversible resolution of an informational superposition. The universe is not a fact — it is a process. It is not a thing — it is an act. And the name of that act is Entropicity.

It means, third, that the deepest question of metaphysics — why there is something rather than nothing — receives a precise and satisfying answer. There is something rather than nothing because Entropicity is irreversible: once a bit is erased, the Landauer energy it releases cannot be un-released, the physical reality it creates cannot be un-created. The existence of the physical world is not a brute fact; it is a thermodynamic necessity. The informational substrate, once it began to resolve, necessarily produced physical reality — and it continues to produce it, moment by moment, at the rate of $kT \ln 2$ per Planck-area bit.

The Theory of Entropicity is, in the deepest sense, a theory of genesis — an account of how the universe comes to be, not once in a distant past, but continuously, in every Planck moment of time, through the primordial creative act of informational foreclosure. The research program it opens is vast and deep. The theoretical tasks of working out the quantum emergence problem, the pre-cosmological initial conditions, and the precise relationship to loop quantum gravity and string theory will occupy theorists for years. The observational tests — gamma-ray dispersion, black hole thermodynamics, galactic rotation — will engage astronomers and experimentalists for decades. But the foundation has been laid. The theory has been stated. The program has begun.

Physics has always advanced by discovering that what appeared to be the fundamental substance of reality — matter, field, spacetime — is in fact emergent from something deeper. The Theory of Entropicity proposes the deepest emergence of all: that physical reality itself — in all its richness and complexity, from the quantum vacuum to the large-scale structure of the cosmos — is the ongoing thermodynamic consequence of an informational universe resolving itself into existence, one bit at a time.

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*Obidi's Theory of Entropicity — Monograph Version 1.0 / INFINITE Obidi
/ Independent Theoretical Research / June 2026*

*This document is an original theoretical work. All theoretical constructs,
field equations, and the concept of Entropicity are original contributions of
the author.*